



HINTS & SOLUTIONS

TOPIC : APPLICATION OF DERIVATIVES

EXERCISE # 1

PART-1

Section (A)

A-1. (i) $\frac{dy}{dx} = 6x + 4 \Rightarrow \left. \frac{dy}{dx} \right|_{(0,5)} = 4$

Equation of tangent is $\frac{y-5}{x-0} = 4 \Rightarrow y = 4x + 5$

(ii) $\frac{dy}{dx} \text{ at } (1,1) = - \left(\frac{2x+3y}{3x+2y} \right)_{(1,1)} = -1$

$\Rightarrow$ equation of tangent is $(y-1) = -(x-1)$ and equation of normal is $(y-1) = (x-1)$

(iii) At $t = \frac{1}{2}$, the value of $x = \frac{2a}{5}$ and $y = \frac{a}{5}$

Also $\frac{dx}{dt} = \frac{4at}{(1+t^2)^2}$ and $\frac{dy}{dt} = \frac{2at^2(3+t^2)}{(1+t^2)^2}$

$\frac{dy}{dx} = \frac{t}{2}(3+t^2) \Rightarrow \left. \frac{dy}{dx} \right|_{t=\frac{1}{2}} = \frac{13}{16}$

equation of tangent is $\left(y - \frac{a}{5} \right) = \frac{13}{16} \left(x - \frac{2a}{5} \right) \Rightarrow 13x - 16y = 2a$

equation of normal is $\left(y - \frac{a}{5} \right) \frac{13}{16} + x - \frac{2a}{5} = 0 \Rightarrow 16x + 13y = 9a$

(iv) $\frac{dy}{dx} \text{ at } (0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin 1/h - 0}{h}$
 $= \lim_{h \rightarrow 0} (h \sin(1/h)) = 0 \Rightarrow$ equation of tangent is $\frac{y-0}{x-0} = 0 \Rightarrow y = 0$

A-2 $y^2 - 2x^2 - 4y + 8 = 0$

$2y \frac{dy}{dx} - 4x - 4 \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = \frac{4x}{2y-4}$

$\left(\frac{dy}{dx} \right)_{(h, k)} = \frac{4h}{2k-4}$

Equation of tangent is $(y-k) = \frac{4h}{(2k-4)} (x-h)$.

It passes through $(1, 2)$

$(2-k)(2k-4) = 4h(1-h)$

or, $4k - k^2 - 4 = 2h - 2h^2$

$\therefore k^2 - 2h^2 - 4k + 8 = 0$

$\therefore 2h - 4 = 0$ or $h = 2 \Rightarrow k = 0$ or 4

Equation of tangent at $(2, 0)$; $y = \frac{8}{(-4)} (x-2)$

or $y = -2x + 4$ or $y + 2x = 4$

and equation of tangent at $(2, 4)$ is $y = 2x$





- (ii) Let point Q be $\left(h, \frac{h^2}{4}\right)$ and point P be the point of contact on the curve.

Now, m_{PQ} = slope of the normal at Q.

Differentiating $x^2 = 4y$ w.r.t we get $2x = 4 \frac{dy}{dx}$ or $\frac{dy}{dx} = \frac{x}{2}$

or Slope of the normals at Q = $\frac{dy}{dx}\bigg|_{x=h} = -\frac{2}{h}$

$$\text{or } \frac{\frac{h^2}{4} - 2}{h - 1} = -\frac{2}{h} \quad [\text{From (1)}]$$

$$\text{or } \frac{h^3}{4} - 2h = -2h + 2 \quad \text{or } h^3 = 8 \quad \text{or } h = 2$$

Hence, the coordinates of point Q are (2, 1) and so the equation of the required normal becomes $x + y = 3$

- A-3. (i) Slope of normal equal to -1.

$$\Rightarrow \frac{dy}{dx} = \frac{x^2}{6y} = 1 \quad x^2 = 6y \quad \text{and } 9y^2 = x^3 \quad \Rightarrow x = 0, 4 \Rightarrow \text{point is } \left(4, \frac{8}{3}\right)$$

- (ii) Differentiating equation of curve w.r.t. x, $2y \frac{dy}{dx} = (2-x)^2 + 2x(2-x)(-1) \frac{dy}{dx}\bigg|_{(1,1)} = \frac{1+(-2)}{2} = -\frac{1}{2}$

$$\text{Equation of tangent is } (y - 1) = -\frac{1}{2}(x - 1) \quad \text{or} \quad 2y + x = 3.$$

Solving the equations of tangent and curve:

$$\begin{aligned} y^2 &= (-2y + 3)(2 - 3 + 2y)^2 & \text{or} & \quad y^2 = (3 - 2y)(2y - 1)^2 \\ \text{or} \quad y^2 &= (3 - 2y)(4y^2 + 1 - 4y) & \text{or} & \quad y^2 = 12y^2 + 3 - 12y - 8y^3 - 2y + 8y^2 \\ \text{or} \quad 8y^3 - 19y^2 + 14y - 3 &= 0 & \text{or} & \quad (y - 1)(8y^2 - 11y + 3) = 0 \\ \text{or} \quad (y - 1)(8y - 3)(y - 1) &= 0 & \text{or} & \quad y = 1, 3/8 \end{aligned}$$

$$P(9/4, 3/8)$$

- (iii) Differentiating, $25x^4 - 30x^2 + 1 + 2y' = 0$

$$\text{At } P(0, -3), y' = -\frac{1}{2}$$

The normal at P is $y + 3 = 2x$

Eliminating y with the given equation $x(x^2 - 1)^2 = 0 \rightarrow x = 0, 1, -1$

The line is tangent at (1, -1) and (-1, -5).

- A-4. (i) Slope of $y = x^3$ at (h, k) is $3h^2$

Slope of $y = x^3$ at (h, k) is $3h^2$

$$\Rightarrow \text{equation of tangent is } = \frac{y - k}{x - h} = 3h^2$$

$$\Rightarrow y = 3h^2x - 3h^3 + k \Rightarrow y = 3h^2x - 2h^3 \Rightarrow (-2h^3)^2 = 1(3h^2)^2 + 112$$

$$\Rightarrow 4h^6 - 9h^4 - 112 = 0$$

$$(h^2 - 4)(4h^4 + 7h^2 + 28) = 0 \Rightarrow h = 2, -2$$

Tangent are $y = 12x - 16$ or $y = 12x + 16$



(ii) $\frac{dy}{dx}$ for $y = \frac{1}{x^2}$ is $\frac{-2}{x^3}$

$$\left. \frac{dy}{dx} \right|_{(h,x)} = \frac{-2}{h^3}$$

equation of normal is $\frac{y-k}{x-h} = \frac{h^3}{2}$

$$\Rightarrow 2y - \frac{2}{h^2} = h^3x - h^4$$

it passes from $\left(0, \frac{1}{2}\right) \Rightarrow 1 - \frac{2}{h^2} = -h^4 \Rightarrow h^6 + h^2 - 2 = 0$

$$\Rightarrow (h^2 - 1)(h^4 + h^2 + 2) = 0 \Rightarrow h = \pm 1$$

$\Rightarrow$ common normals are $x - 2y + 1 = 0$ or $2y + x - 1 = 0$

A-5. (i) $xy + ax + by = 0$

As (1,1) lies on curve, so

$$1 + a + b = 0 \text{ \& } \theta = \tan^{-1} 2 \quad \text{or} \quad \frac{dy}{dx} = 2$$

Differentiating equation of curve w.r.t. x, $x \frac{dy}{dx} + y + a + b \frac{dy}{dx} = 0$,

put $x = 1, y = 1 \Rightarrow \frac{dy}{dx} = -\frac{(1+a)}{1+b} = 2$

or $1 + a + 2 + 2b = 0$ or $3 + a + 2b = 0 \quad b = -2 \text{ \& } a = 1$

(ii) $(-2, 0)$ lies on both curves $\Rightarrow -8a + 4b - 6 + 5 = 0 \dots(i)$

$\left(\frac{dy}{dx} \text{ at } (-2, 0) \text{ for second curve}\right) = \left(\frac{dy}{dx} \text{ at } (-2, 0) \text{ for first curve}\right) = 0$

$\Rightarrow 12a - 4b + 3 = 0 \dots(ii)$

$\Rightarrow$ Solving (i) and (ii) we get $a = \frac{1}{2}, b = \frac{3}{4}$

Section (B)

B-1. For $C_1, \left. \frac{dy}{dx} \right|_{(1,0)} = \left(\frac{2^x}{x} + \ln x \cdot 2^x \cdot \ln 2 \right)_{(1,0)} = 2$

For $C_2, \left. \frac{dy}{dx} \right|_{(1,0)} = \left(x^{2x} \cdot \ln x \times 2 + 2x(x)^{2x-1} \right)_{(1,0)} = 2 \Rightarrow \theta = 0 \Rightarrow \cos \theta = 1$

B-2. For $C_1, \left. \frac{dy}{dx} \right|_{x=1} = \left(\frac{1}{x} \right)_{x=1} = 1$

For $C_1, \left. \frac{dy}{dx} \right|_{x=e} = \left(\frac{1}{x} \right)_{x=e} = \frac{1}{e}$

For $C_2, \left. \frac{dy}{dx} \right|_{x=1} = \left(\frac{2 \ln x}{x} \right)_{x=1} = 0$

For $C_2, \left. \frac{dy}{dx} \right|_{x=e} = \left(\frac{2 \ln x}{x} \right)_{x=e} = \frac{2}{e}$

angle between curves at (1,0) is $\pi/4$ and angle between curves at (e,1) is $\tan^{-1} \left(\frac{e}{e^2 + 2} \right)$





B-3. Given curves are $y^2 = 4x + 4$ and $y^2 = 36(9 - x)$ (i)

On solving, we get the point (8, 6) and (8, -6)

On differentiating equation (i), we get

$$2y \frac{dy}{dx} = 4 \text{ and } 2y \frac{dy}{dx} = -36$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{y} \text{ and } \frac{dy}{dx} = \frac{-18}{y}$$

$$\text{At point (8, 6), } m_1 = \frac{dy}{dx} = \frac{1}{3} \text{ and } m_2 = \frac{dy}{dx} = -3$$

$$m_1 m_2 = -1$$

B-4. $ax^2 + by^2 = 1 \Rightarrow \frac{dy}{dx} = \frac{-ax}{by}$

$$Ax^2 + By^2 = 1 \Rightarrow \frac{dy}{dx} = \frac{-Ax}{By}$$

Product of slopes = -1

$$\Rightarrow aAx^2 + bBy^2 = 0$$

$$\text{Eliminating } x^2, y^2 \Rightarrow \begin{vmatrix} a & b & 1 \\ A & B & 1 \\ aA & bB & 0 \end{vmatrix} = 0$$

$$\Rightarrow (AbB - aAB) - (abB - abA) = 0$$

$$AB(b - a) - ab(B - A) = 0$$

$$\Rightarrow ab(A - B) = AB(a - b)$$

B-5. Let C_1 is $y = x - 2$ and C_2 is $y = x^2 + 3x + 2$
Slope of common normal is -1 (if possible)

$$\text{Now, for } C_2, \frac{dy}{dx} = 2x + 3 = 1 \Rightarrow x = -1$$

$$\Rightarrow \text{Point on } C_2 \text{ where normal has slope equal to } -1 \text{ is } (-1, 0)$$

$$\Rightarrow \text{Shortest distance between } C_1 \text{ and } C_2 \text{ is distance of } (-1, 0) \text{ from } y = x - 2 \text{ which is } \frac{3}{\sqrt{2}}$$

B-6. Equation of normal to $y^2 = 4x$ is $y = -tx + 2t + t^3$ at point $(t^2, 2t)$. If it is common normal to $(x - 6)^2 + y^2 = 1$ then (6, 0) satisfies the above equation of normal

$$\Rightarrow t^3 - 4t = 0 \Rightarrow t = 0, 2, -2$$

$$\Rightarrow \text{feet of these common normal are } (0, 0), (4, 4), (4, -4)$$

$$\Rightarrow \text{Distance of these feet from } (6, 0) \text{ are } 6, \sqrt{20}, \sqrt{20} \text{ respectively}$$

$$\Rightarrow \text{shortest distance between } y^2 = 4x \text{ and } (x - 6)^2 + y^2 = 1 \text{ is } \sqrt{20} - 1$$

Section (C)

C-1. (i) Let P be perimeter

$$P = 2x + 2y$$

$$\frac{dP}{dt} = 2 \frac{dx}{dt} + 2 \frac{dy}{dt}$$

$$\frac{dA}{dt} = \frac{dx}{dt} y + x \frac{dy}{dt} - 6 + 4 = -2$$

(ii) Let A be area $A = xy$

$$\frac{dA}{dt} = \frac{dx}{dt} y + x \frac{dy}{dt} = -18 + 20 = 2$$

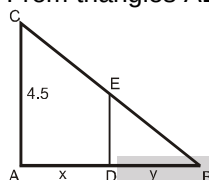


C-2. We have to obtain $\frac{d(y^2)}{d(x^2)} = \frac{2y \frac{dy}{dx}}{2x} = \frac{y}{x} \cdot \frac{dy}{dx}$

$$y = x - x^2 \Rightarrow \frac{dy}{dx} = 1 - 2x$$

$$\frac{d(y^2)}{d(x^2)} = \frac{y}{x} (1 - 2x) = \frac{(x - x^2)(1 - 2x)}{x} = 2x^2 - 3x + 1$$

C-3. Let AC be pole, DE be man and B be farther end of shadow as shown in figure
From triangles ABC and DBE



$$\frac{4.5}{x+y} = \frac{1.5}{y}$$

$$3y = 1.5x$$

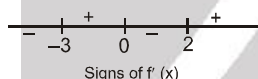
$$\frac{dy}{dt} = 2, \quad \frac{d}{dt} (x+y) = \frac{dx}{dt} + \frac{dy}{dt}$$

C-4. $V = a^3 \Rightarrow \Delta V = \frac{dV}{da} \Delta a = 3a^2 \Delta a = 3a^2 \times \frac{2a}{100} = \frac{6 \times 5^3}{100} = 7.5 \text{ m}^3$

Section (D)

D-1. $f'(x) = \frac{2+x-2\sqrt{1+x}}{2(x+1)^{3/2}} = \frac{(\sqrt{x+1}-1)^2}{2(x+1)^{3/2}} \geq 0, x > -1 \Rightarrow f(x) \text{ is increasing.}$

D-2. (i) Let $f(x) = \frac{x^4}{4} + \frac{x^3}{3} - 3x^2 + 5$



$$f'(x) = x(x+3)(x-2)$$

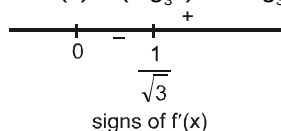
M.D. in $(-\infty, -3]$

M.I. in $[-3, 0]$

M.D. in $[0, 2]$

M.I. in $[2, \infty)$

(ii) Let $f(x) = (\log_3 x)^2 + \log_3 x$



$$= \frac{(\ln x)^2}{(\ln 3)^2} + \frac{\ln x}{\ln 3} \quad f'(x) = \frac{2(\ln x) + \ln 3}{x(\ln 3)^2}$$

M.D. in $\left(0, \frac{1}{\sqrt{3}}\right]$

M.I. in $\left[\frac{1}{\sqrt{3}}, \infty\right)$





D-3. $g(x)$ is monotonically increasing

$$\Rightarrow g'(x) \geq 0 \text{ \& } f(x) \text{ is M.D.} \Rightarrow f'(x) \leq 0$$

$$\frac{d}{dx} (f \circ g)(x) = \frac{d}{dx} (f(g(x))) = f'(g(x)) \cdot g'(x) \leq 0$$

$$\text{as } f'(x) \leq 0 \text{ \& } g'(x) \geq 0$$

$$\Rightarrow (f \circ g)(x) \text{ is monotonically decreasing} \Rightarrow f'(x) \leq 0$$

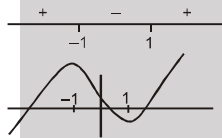
$$\text{Also } x+1 > x-1$$

$$\Rightarrow f(x+1) < f(x-1) \quad \text{as } f(x) \text{ is M.D.}$$

$$\Rightarrow g(f(x+1)) < g(f(x-1)) \quad \text{as } g(x) \text{ is M.I.}$$

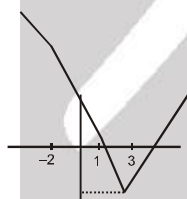
D-4. $f'(x) = \begin{cases} a & ; x < 0, \\ 2x & ; x > 0. \end{cases}$
 $f'(x) > 0 \Rightarrow a > 0$

D-5. (i) $f(x) = 3x^2 - 3$
 $= 3(x-1)(x+1)$



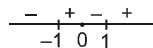
at $x = 1$ point of minima
 $\therefore$ Neither increasing nor decreasing
 at $x = 2$ increasing

(ii) at $x = -2$ decreasing
 at $x = 0$ decreasing
 at $x = 3$ neither increasing nor decreasing
 at $x = 5$ increasing



(iii) $f'(x) = \frac{1}{3x^{2/3}} \Rightarrow f'(x) > 0 \forall x \in \mathbb{R} - \{0\}$
 $\Rightarrow$ strictly increasing at $x = 0$

(iv) $f'(x) = \frac{2(x^2+1)(x-1)(x+1)}{x^3} \Rightarrow$ strictly increasing at $x = 2$, neither I nor D at $x = 1$



(v) $f'(x) = \begin{cases} 3x^2 + 4x + 5, & x < 0 \\ 3\cos x, & x > 0 \end{cases} \Rightarrow f'(0^-) = +5$
 and $f'(0^+) = 3 \Rightarrow$ strictly increasing at $x = 0$



D-6. Let $f(x) = \frac{\sin x}{x} \Rightarrow f'(x) = \frac{x \cos x - \sin x}{x^2} = \frac{\cos x(x - \tan x)}{x^2} < 0 \forall x \in \left(0, \frac{\pi}{2}\right)$

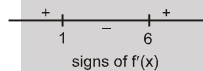
$\Rightarrow f(x)$ is decreasing in $\left(0, \frac{\pi}{2}\right) \Rightarrow f\left(\frac{1}{10}\right) > f\left(\frac{1}{9}\right) \Rightarrow \left(\frac{\sin\left(\frac{1}{10}\right)}{\frac{1}{10}}\right) > \left(\frac{\sin\left(\frac{1}{9}\right)}{\frac{1}{9}}\right)$

D-7. Let $h(x) = f(x) - g(x)$
 $h(0) = f(0) - g(0) \Rightarrow h(0) = 0$
 $h'(x) = f'(x) - g'(x) \leq 0$ for $x \geq 0$
 $\Rightarrow h(x)$ is decreasing for $x \geq 0$
 $x \geq 0$
 $h(x) \leq h(0)$
 $h(x) \leq 0$
 $f(x) - g(x) \leq 0$
 $f(x) \leq g(x)$

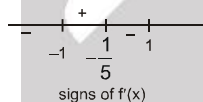
D-8. $f'(x) = \begin{cases} -1 & , \quad 0 < x < 1 \\ 2x & x > 1 \end{cases}$

$f'(x)$ changes sign from negative to positive.
 $f(1^-) = 2, f(1^+) = 1 + \ln b$ and $f(1) = 1 + \ln b$.
 $f(1^-) \geq f(1) \Rightarrow 2 \geq 1 + \ln b$
 $\Rightarrow \ln b \leq 1 \quad 0 < b \leq e$

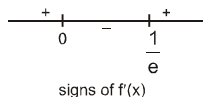
D-9. (i) $f'(x) = 6(x-1)(x-6)$
 Local maxima at $x = 1$
 Local minima at $x = 6$



(ii) $f'(x) = -(x-1)^2(x+1)(5x+1)$
 Local minima at $x = -1$
 Local maxima at $x = -\frac{1}{5}$



Neither local minima nor local maxima at $x = 1$.



(iii) $f'(x) = \ln x + 1$
 Local minima at $x = \frac{1}{e}$

No local maxima

D-10. (i) $g(t) = (t-1)(t-2)^2$
 $g'(t) = (3t-4)(t-2)$



$$\begin{array}{c} + \quad - \quad + \\ \hline 4/3 \quad 2 \end{array}$$

local maxima at $x = \log_2 \frac{4}{3}$ and local minima at $x = 1$

(ii) $f'(x) = xe^{-x}(2-x)$

$\Rightarrow$ local min at 0, local max at 2

$$\begin{array}{c} - \quad + \quad - \\ \hline 0 \quad 2 \end{array}$$

(iii) $f'(x) = -3\sin 2x (2\cos x + 1) (\cos x + 2)$

local max at $x = 0, \frac{2\pi}{3}$

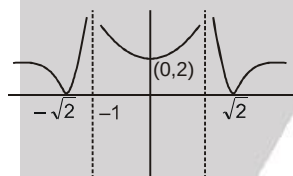
local min at $x = \frac{\pi}{2}, \pi$

$$\begin{array}{c} - \quad + \quad - \\ \hline 0 \quad \pi/2 \quad 2\pi/3 \quad \pi \end{array}$$

(iv) $f'(x) = \frac{2(1+x^{1/3})}{x^{1/3}}$

$$\begin{array}{c} + \quad - \quad + \\ \hline -1 \quad 0 \end{array}$$

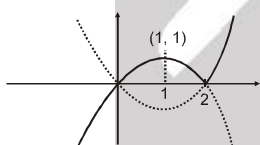
local maxima at -1 and local minima at 0



(v)

local minima at $x = \pm\sqrt{2}, 0$

D-11. $f(x) = \begin{cases} -x^2 + 2x & : x < 2 \\ x^2 - 2x & : x \geq 2 \end{cases}$



Graph of $y = f(x)$

Section (E)

E-1. (i) $f'(x) = 3x^3$

$f'(x) = 0 \Rightarrow x = 0$

$x = -2, f(-2) = -8$

$x = 0, f(0) = 0$

$x = 2, f(2) = 8$

Minimum = -8 , maximum = 8

(ii) $f'(x) = \cos x - \sin x$



$$f'(x) = 0 \Rightarrow x = \frac{\pi}{4}$$

$$x = 0, \quad f(0) = 1$$

$$x = \frac{\pi}{4}, \quad f\left(\frac{\pi}{4}\right) = \sqrt{2}$$

$$x = \pi, \quad f(\pi) = -1$$

$$\text{Minimum} = -1, \text{Maximum} = \sqrt{2}$$

(iii) $f'(x) = 4 - x$

$$f'(x) = 0 \Rightarrow x = 4$$

$$x = -2, \quad f(-2) = -10$$

$$x = 4, \quad f(4) = 8$$

$$x = \frac{9}{2}, \quad f\left(\frac{9}{2}\right) = \frac{63}{8}$$

$$\text{Minimum} = -10, \text{Maximum} = 8$$

(iv) $f'(x) = \cos x - \sin 2x$

$$f'(x) = 0 \Rightarrow \cos x = 0, \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{2}, x = \frac{\pi}{6}$$

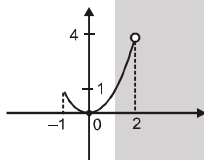
$$x = 0, f(0) = \frac{1}{2}$$

$$x = \frac{\pi}{6}, f\left(\frac{\pi}{6}\right) = \frac{3}{4}$$

$$x = \frac{\pi}{2}, f\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

$$\text{Minimum} = \frac{1}{2}, \text{Maximum} = \frac{3}{4}$$

E-2.



Graph of $f(x)$

$x = 0$ is local minima and global maximum is not defined

E-3. Let No. of children of John & Angina = y

$$\therefore x + (x + 1) + y = 24$$

$$y = 23 - 2x$$

Number of fights

$$F = x(x + 1) + x(23 - 2x) + (x + 1)(23 - 2x)$$

$$F = -3x^2 + 45x + 23$$

$$\frac{dF}{dx} = 0 \Rightarrow -6x + 45 = 0 \Rightarrow x = 7.5$$

But ' x ' will be integral.

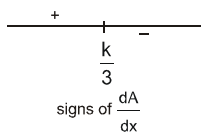
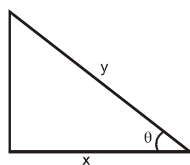
check $x = 6$ or $x = 7$

$$F = 191$$





E-4. $x + y = k$ constant $x + y = k$

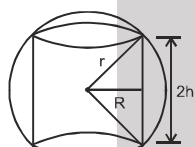


Area of triangle be A .

$$A = \frac{1}{2} x \sqrt{y^2 - x^2} \quad A = \frac{1}{2} x \sqrt{k} \sqrt{k-2x} \quad \frac{dA}{dx} = \frac{\sqrt{k} (k-3x)}{2 \sqrt{k-2x}} \quad \text{Area is maximum when } x = \frac{k}{3}.$$

$$\Rightarrow y = \frac{2k}{3} \Rightarrow \cos \theta = \frac{\frac{k}{3}}{\frac{2k}{3}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}.$$

E-5. $R^2 + r^2 = h^2$
 $R^2 = h^2 - r^2$



Figure

volume of cylinder,

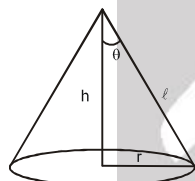
$$V = \pi R^2 (2h) = \pi (2h) (\sqrt{r^2 - h^2})^2$$

$$\frac{dV}{dh} = 2\pi (r^2 - h^2) + 2\pi h(-2h) = 0$$

$$\Rightarrow r^2 = 3h^2 \Rightarrow h = \frac{r}{\sqrt{3}}$$

$$\frac{d^2V}{dh^2} < 0 \text{ at } h = \frac{r}{\sqrt{3}} \Rightarrow V_{\max} = 2\pi \frac{r}{\sqrt{3}} \left(r^2 - \frac{r^2}{3} \right) = \frac{4\pi r^3}{3\sqrt{3}}$$

E-6. $h = \ell \cos \theta$



Figure

$$r = \ell \sin \theta$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \ell^3 \sin^2 \theta \cos \theta$$

$$\frac{dV}{d\theta} = \frac{1}{3} \pi \ell^3 (2 \sin \theta \cos^2 \theta - \sin^3 \theta) \quad \frac{dV}{d\theta} = \frac{1}{3} \pi \ell^3 \sin \theta (2 - 3 \sin^2 \theta) = 0 \text{ at}$$

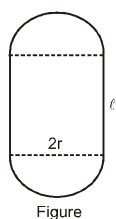
$$\sin \theta = \sqrt{\frac{2}{3}}$$

$$\Rightarrow \tan \theta = \sqrt{2}$$





E-7. $2\ell + 2\pi r = 440$

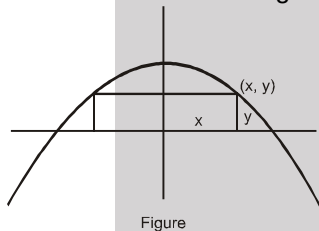


$$A = \ell 2r = -2\pi r^2 + 440r$$

$$\frac{dA}{dr} = -4\pi r + 440 = 0$$

at $r = \frac{110}{\pi}$

E-8. Let A be area of rectangle.



$$A = (2x)(y) = 2x(12 - x^2)$$

$$\frac{dA}{dx} = 6(2 - x)(2 + x)$$

At $x = 2$, A has largest value. Largest A = 32

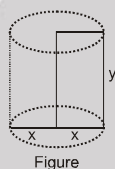
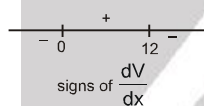
E-9. Let x, y be dimensions of rectangle.

$$\Rightarrow 2x + 2y = 36.$$

Let V be volume swept

$$V = \pi x^2 y$$

$$V = \pi x^2(18 - x)$$



$$\frac{dV}{dx} = \pi x \cdot 3(12 - x)$$

At $x = 12$, V has maximum value. $\Rightarrow y = 6$

E-10. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{C - R_1}$

$$R = R_1(C - R_1)/C$$

$$\frac{dR}{dR_1} = \frac{C - 2R_1}{C}$$

$$\frac{dR}{dR_1} = 0 \text{ at } R_1 = \frac{C}{2}$$

$$R_2 = \frac{C}{2}$$

So $R_1 = R_2$



Section (F)

F-1. $f(x) \downarrow, g(x) \uparrow$

$$\text{let } h(x) = f(x) - g(x)$$

$$h(1) = 2, h(2) = -6$$

$$\Rightarrow h(x) = 0 \text{ has at least one root in } (1, 2).$$

$$\text{but } h'(x) = f'(x) - g'(x) < 0$$

$$\Rightarrow h(x) \text{ is always decreasing so only one root}$$

F-2. $f(x)$ is continuous on $[a, b]$, $f(x)$ is

differentiable on (a, b) , $f(a) = p = f(b)$.

Conditions in Rolle's theorem are satisfied.

$$f'(x) = \left(\frac{x(a+b)}{x^2+ab} \right) \left(1 - \frac{ab}{x^2} \right) \frac{1}{(a+b)}$$

$$f'(x) = 0 \Rightarrow x = \sqrt{ab} \in (a, b)$$

$$\text{i.e. } f'(c) = 0, c = \sqrt{ab}$$

conclusion of Rolle's theorem also valid.

F-3. Let $f'(x) = 3x^2 + px - 1 \Rightarrow f(x) = x^3 + \frac{px^2}{2} - x + c$

$$f(-1) = \frac{p}{2} + c = f(1)$$

$$\Rightarrow f(x) \text{ satisfies conditions in Rolle's theorem } \Rightarrow f'(c) = 0 \text{ for at least one } c \in (-1, 1)$$

$$\Rightarrow 3x^2 + px - 1 = 0 \text{ has at least one root in } (-1, 1).$$

F-4. $f(x) = 0 \Rightarrow \sin \frac{\pi}{x} = 0 \Rightarrow \frac{\pi}{x} = n\pi$

$$\Rightarrow x = \frac{1}{n}, n \in \mathbb{N}$$

$$x = \dots, \frac{1}{n}, \frac{1}{n-1}, \dots, \frac{1}{3}, \frac{1}{2}, 1.$$

$$\text{Consider interval } \left[\frac{1}{n+1}, \frac{1}{n} \right] f\left(\frac{1}{n+1}\right) = 0 = f\left(\frac{1}{n}\right)$$

$$\text{By Rolle's theorem } f'(x) \text{ vanishes at least once in } \left(\frac{1}{n+1}, \frac{1}{n} \right)$$

Infinite number of such intervals are there. Hence $f'(x)$ vanishes at infinite number of points in $(0, 1)$

F-5. Let $h(x) = f(x)g'(x)$ $h(a) = 0 = h(b)$

By Rolle's theorem on $[a, b]$ $h'(x) = 0$, for at least one $c \in (a, b)$. $\Rightarrow f'(c)g'(c) + f(c)g''(c) = 0$

F-6. $f'(c) = \frac{f\left(\frac{\pi}{5}\right) - f(0)}{\frac{\pi}{5} - 0}$; $f'(c) = \sec^2 c$ which is strictly increasing $f'(c) = \frac{f\left(\frac{\pi}{5}\right) - f(0)}{\frac{\pi}{5} - 0}$; $f'(c) = \sec^2 c$

$$f'(0) < f'(c) < f'\left(\frac{\pi}{5}\right) \sec^2 0 < \frac{\tan \frac{\pi}{5}}{\frac{\pi}{5}} < \sec^2 \frac{\pi}{5} < \sec^2 \frac{\pi}{4} \frac{\pi}{5} < \tan \frac{\pi}{5} < \frac{2\pi}{5}$$



F-7. Let $\phi(x) = f(x) - 2g(x)$; $x \in [0, 23] \Rightarrow \phi'(x) = f'(x) - 2g'(x)$

Also $\phi(0) = f(0) - 2g(0) = 2 - 0 = 2$

$\phi(23) = f(23) - 2g(23) = 22 - 20 = 2$

Since $f(x)$ and $g(x)$ are differentiable in $[0, 23]$ hence $\phi(x)$ is also continuous in $[0, 23]$ and differentiable in $[0, 23]$, so all the conditions of Rolle's theorem are satisfied. Hence there exist a number c , $0 < c < 23$ for which $\phi'(c) = 0$

F-8. $f(a) = f(b) = 0$ and we know $\sin^3 x$, xe^x , $\frac{x}{1+x^2}$ are continuous and differentiable for $x \in \mathbb{R}$ therefore $f(x)$ is also continuous and differentiable in $[a, b]$ hence by Rolle's theorem there exist a real number $c \in [a, b]$ such that $f'(c) = 0$.

F-9. $f(0) = 0$ and $f(6) = 2$
so $f(0) \neq f(6)$
 $f(x)$ is discontinuous at $x = 4$ and nondifferentiable at $x = 1, 4$ but $f'(3) = 0$

F-10. Let $g(x) = \frac{f(x)}{x}$, $x \in [a, b]$. By Rolle's theorem $g'(x_0) = 0 \Rightarrow \frac{f'(x_0)x_0 - 1.f(x_0)}{x_0^2} = 0 \Rightarrow f'(x_0) = \frac{f(x_0)}{x_0}$

PART - II

Section (A)

A-1. $f'(0) = \lim_{x \rightarrow 0} \frac{\sin x^2 - 0}{x - 0} = 1$ (slope of tangent)
slope of normal is -1
Equation of normal is $y - 0 = -(x - 0)$

A-2. $\frac{dy}{dx} = -\frac{1}{2\sqrt{x}} \Rightarrow \frac{dy}{dx}\bigg|_{(1,1)} = -\frac{1}{2}$
Slope of normal = 2
Equation of normal is $2x - y = 1$

A-3. $\frac{dy}{dx}\bigg|_{(3,0)}$ is 4 $\Rightarrow \theta = \tan^{-1} 4$

A-4. $\frac{dy}{dx} = 5x^4$

Equation of tangent at (h, k) is $\frac{y-k}{x-h} = 5h^4$ Which passes through $(2, 2)$

$\Rightarrow (2 - k) = 5h^4 (2 - h) \Rightarrow -h^5 = 10h^4 - 5h^5 \Rightarrow 4h^5 = 10h^4 \Rightarrow h = 0, \frac{5}{2}$

$\Rightarrow$ Equation of tangents are $y - 2 = 0$ and $16(y - 2) = 5^5(x - 2)$

A-5. Equation of tangent is
 $y - 4/h = -4/h^2 (x - h)$
It passes through $(0, 1) \Rightarrow 1 - 4/h^2 = 4h/h^2$
 $\Rightarrow h - 4 = 4 \Rightarrow h = 8 \Rightarrow$ tangent is $y - 1/2 = -1/16 (x - 8)$.



A-6. $y - e^{xy} + x = 0$

Differentiating w.r.t. to y

$$1 - e^{xy} \left(\frac{dx}{dy} \cdot y + x \right) + \frac{dx}{dy} = 0$$

$$\frac{dx}{dy} = 0$$

$$1 - xe^{xy} = 0$$

$$xe^{xy} = 1 \quad x = 1, y = 0$$

Point is (1, 0)

A-7. $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a(-\sin\theta)}{a(1+\cos\theta)}$

$$\left. \frac{dy}{dx} \right|_{\theta=\frac{\pi}{3}} = \frac{-\frac{\sqrt{3}}{2}}{\frac{3}{2}} = -\frac{1}{\sqrt{3}}$$

$$\tan \alpha = -\frac{1}{\sqrt{3}} \Rightarrow \alpha = \pi - \frac{\pi}{6} \Rightarrow \alpha = \frac{5\pi}{6}$$

A-8. $\left. \frac{dy}{dx} \right|_{(3t, 4/t)} = \left. \frac{-12}{x^2} \right|_{(3t, 4/t)} = \frac{-4}{3t^2}$

equation of normal $\frac{y - 4/t}{x - 3t} = \frac{3t^2}{4}$ which passes through $(3t_1, 4/t_1)$

$$\Rightarrow \frac{4}{t_1} - \frac{4}{t} = \frac{3t^2}{4} (3t_1 - 3t) \Rightarrow \frac{4}{tt_1} = \frac{-9t^2}{4} \Rightarrow t_1 = \frac{-16}{9t^3}$$

A-9. $\frac{dy}{dx}$ for $y = x^2 + \frac{1}{x}$ is $2x - \frac{1}{x^2}$

$$\frac{dy}{dx} \text{ at } (h, k) \text{ is } 2h - \frac{1}{h^2}$$

equation of tangent at (h, k) is $\frac{y - (h^2 + 1/h)}{x - h} = 2h - \frac{1}{h^2}$

$$y = \left(2h - \frac{1}{h^2} \right) x - 2h^2 + \frac{1}{h} + h^2 + \frac{1}{h}$$

$$y = \left(2h - \frac{1}{h^2} \right) x + \frac{2}{h} - h^2 \Rightarrow \frac{2}{h} - h^2 = \frac{1}{2h - 1/h^2} \Rightarrow \frac{2 - h^3}{h} = \frac{h^2}{2h^3 - 1} \Rightarrow h^6 - 2h^3 + 1 = 0$$

$$\Rightarrow (h^3 - 1)^2 = 0 \Rightarrow (h - 1)^2 (h^2 + h + 1)^2 = 0 \Rightarrow h = 1 \Rightarrow \text{equation of tangent is } y = x + 1$$

A-10. $1 = 1 + b + c \Rightarrow c = -b$

$$\frac{dy}{dx} = 2x + b = 2 + b \text{ at } (1, 1)$$

equation of tangent is $y - 1 = (2 + b)(x - 1) \Rightarrow x \text{ intercept} = \frac{b+1}{b+2}$

$$y \text{ intercept} = -(b+1) \Rightarrow \left| \frac{1}{2} \times \left(\frac{b+1}{b+2} \right) \times (b+1) \right| = 2$$

$$b^2 + 2b + 1 = 4b + 8 \text{ on } -4b - 8$$

$$b^2 - 2b - 7 = 0 \text{ on } b^2 + 6b + 9 = 0 \Rightarrow \text{integral } b = -3$$



Section (B)

B-1. For C_1 , $\left. \frac{dy}{dx} \right|_{(0,1)} = a^x \ln a \Big|_{(0,1)} = \ln a$

For C_2 , $\left. \frac{dy}{dx} \right|_{(0,1)} = b^x \ln b \Big|_{(0,1)} = \ln b \Rightarrow$ angle between curve is $\left| \frac{\log a - \log b}{1 + \log a \log b} \right| = \left| \frac{\log(a/b)}{1 + \log a \log b} \right|$

B-2. Both curves are confocal, where focus of both curves is $(\sqrt{24}, 0)$ and $(-\sqrt{24}, 0)$

$\Rightarrow$ angle between curves $x^2 + 4y^2 = 32$ and $x^2 - y^2 = 12$ is $\frac{\pi}{2}$

B-3. $\left. \frac{dy}{dx} \right|_{C_1} = \frac{x^2 - y^2}{2xy}$, $\left. \frac{dy}{dx} \right|_{C_2} = \frac{-2xy}{x^2 - y^2} \Rightarrow \left. \frac{dy}{dx} \right|_{C_1} \times \left. \frac{dy}{dx} \right|_{C_2} = -1$

B-4. For C_1 , $\frac{dy}{dx} = \frac{-4x}{a^2 y}$ and For C_2 , $\frac{dy}{dx} = \frac{16}{3y^2} \Rightarrow \frac{-4x}{a^2 y} \times \frac{16}{3y^2} = -1 \Rightarrow \left(\frac{4}{3a^2} \right) \times \left(\frac{16x}{y^3} \right) = -1 \Rightarrow a^2 = 4/3$

B-5. Equation of normal to the curve $y^2 = 8x$ and $y^2 = 4(x-3)$ are $y = mx - 4m - 2m^3$ and $y = m(x-3) - 2m - m^3$ respectively.

$\Rightarrow -4m - 2m^3 = -3m - 2m - m^3 \Rightarrow m^3 - m = 0 \Rightarrow m = 0, 1, -1$

$\Rightarrow$ feet of common normal with slope equal to -1 on the curves $y^2 = 8x$ and $y^2 = 4(x-3)$ are $(4, 2)$ and $(2, 4)$ respectively

Now, distance between points $(4, 2)$ and $(2, 4)$ is $2\sqrt{2}$

B-6. Equation of normal at $\frac{x^2}{32} + \frac{y^2}{18} = 1$ at (h, k) is $\frac{32x}{h} - \frac{18y}{k} = 14$ which passes through $\left(\frac{7}{4}, 0 \right)$

$\Rightarrow \frac{56}{h} - 0 = 14 \quad \Rightarrow h = 4 \quad \Rightarrow k = 3$

Now, distance between $(4, 3)$ and $\left(\frac{7}{4}, 0 \right)$ is $\sqrt{\left(\frac{9}{4} \right)^2 + 9} = \frac{15}{4}$

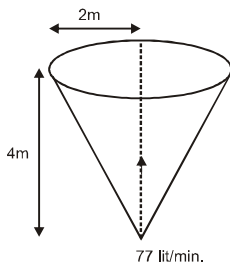
$\Rightarrow$ shortest distance between curves $\frac{x^2}{32} + \frac{y^2}{18} = 1$ and $\left(x - \frac{7}{4} \right)^2 + y^2 = 1$ is $\frac{11}{4}$

Section (C)

C-1. $V = \frac{1}{3} \pi r^2 h \quad \left(\because \frac{r}{h} = \frac{2}{4} = \frac{1}{2} \right)$

$V = \frac{1}{3} \pi \frac{h^3}{4} = \frac{\pi}{12} h^3$

$77 \times 10^3 = \frac{22}{7} \times \frac{1}{4} \times 70 \times 70 \times \frac{dh}{dt} \quad (\because 1 \text{ litre} = 10^3 \text{ c.c.})$



$\therefore \frac{dh}{dt} = 20 \text{ cm/min.}$

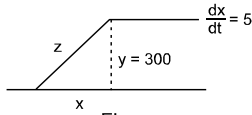


C-2 $x^3 = 12y$

$$3x^2 \frac{dx}{dt} = 12 \frac{dy}{dt} \Rightarrow \frac{dx}{dt} > \frac{dy}{dt}$$

$$\Rightarrow 12 \frac{dy}{dt} \cdot \frac{1}{3x^2} > \frac{dy}{dt}$$

$$\Rightarrow x^2 < 4 \Rightarrow x \in (-2, 2)$$



C-3.

Figure

From figure $z^2 = x^2 + y^2$

$$z \frac{dz}{dt} = x \frac{dx}{dt}$$

$$\text{If } z = 500 \text{ then } x = 400 \Rightarrow 500 \frac{dz}{dt} = 400(5) \Rightarrow \frac{dz}{dt} = 4$$

C-4.

$$\text{Let } y = \tan x \Delta y = \tan(x + \Delta x) - \tan x \Rightarrow \frac{dy}{dx} \Delta x = \tan(x + \Delta x) - \tan x$$

$$\Rightarrow (\sec^2 x) \Delta x = \tan(x + \Delta x) - \tan x$$

$$\text{put } x = 45^\circ, \Delta x = 1^\circ \Rightarrow \frac{2\pi}{180} = \tan 46^\circ - 1 \Rightarrow \tan 46^\circ = 1 + \frac{\pi}{90}$$

C-5.

$$V = 4 \frac{\pi}{3} (10+r)^3, \quad 0 \leq r \leq 15$$

$$\therefore \frac{dV}{dt} = -50.$$

$$4\pi (10+r)^2 \frac{dr}{dt} = -50 \Rightarrow \frac{dr}{dt} = \frac{-1}{18\pi} \text{ (where } r = 5)$$

Section (D)

D-1.

$$f'(x) = 3(a+2)x^2 - 6ax + 9a \leq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a+2 < 0 \quad \text{and} \quad D \leq 0$$

$$\Rightarrow a < -2 \quad \text{and} \quad a \in (-\infty, -3] \cup [0, \infty)$$

$$\Rightarrow a \in (-\infty, -3]$$

D-2.

$$f'(x) = 3x^2 + 2ax + b + 5 \sin 2x \geq 0 \quad x \in \mathbb{R}$$

$$\therefore \sin 2x \geq -1$$

$$\Rightarrow f'(x) \geq 3x^2 + 2ax + b - 5 \quad x \in \mathbb{R} \Rightarrow 3x^2 + 2ax + b - 5 \geq 0 \quad x \in \mathbb{R}$$

$$\Rightarrow 4a^2 - 4 \cdot 3 \cdot (b-5) \leq 0$$

$$\Rightarrow a^2 - 3b + 15 \leq 0$$

D-3.

$$f(x) = \begin{cases} \frac{1-x}{x^2} & x < 1, \quad x \neq 0 \\ \frac{x-1}{x^2}, & x \geq 1 \end{cases}$$

The given function is not differentiable at $x = 1$

$$f'(x) = \begin{cases} \frac{1}{x^2} - \frac{2}{x^3}, & x < 1, \quad x \neq 0 \\ \frac{2}{x^3} - \frac{1}{x^2}, & x > 1 \end{cases} \quad \text{Now } f'(x) < 0 \Rightarrow \begin{cases} \frac{x-2}{x^3} < 0 \text{ given } x < 1 \\ \frac{2-x}{x^3} < 0 \text{ when } x > 1 \end{cases}$$

$f(x)$ decreasing $\forall x \in (0, 1) \cup (2, \infty)$ and $f(x)$ increases $\forall x \in (-\infty, 0) \cup (1, 2)$

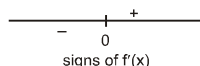
here $f(x)$ is decreasing at all points in $x \in (0, 1) \cup (2, \infty)$ so will also be decreasing at $x = 3$ at $x = 1$ minima and at $x = 2$ maxima



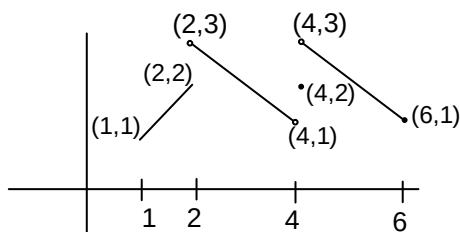


D-4. $f'(x) = (2^2 + 4^2x^2 + 6^2x^4 + \dots + 100^2x^{98})x$

Minimum at $x = 0$

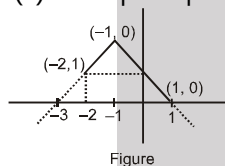


D-6.



Section (E)

E-1. $f(x) = 2 - |x + 1|$



From figure it is clear that greatest, least values are respectively 2, 0

E-2. Since coefficient of x^2 is (+ve)

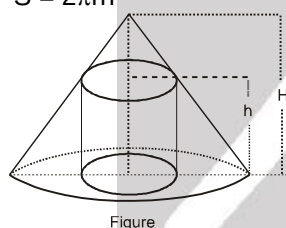
$$\Rightarrow m(b) = -\frac{D}{4a} \quad m(b) = -\frac{(4b^2 - 4(1+b^2))}{4(1+b^2)} \Rightarrow m(b) = \frac{1}{1+b^2}$$

$$\Rightarrow b^2 \geq 0 \Rightarrow 1+b^2 \geq 1 \Rightarrow 0 < \frac{1}{1+b^2} \leq 1 \Rightarrow m(b) \in (0, 1]$$

E-3.

$$\frac{H}{R} = \frac{H-h}{r}$$

$$S = 2\pi rh$$



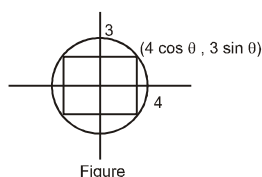
$$= 2\pi H \left(r - \frac{r^2}{R} \right) \frac{dS}{dr} = 2\pi H \left(1 - \frac{2r}{R} \right)$$

$$\text{Maximum at } r = \frac{R}{2}$$

E-4.

$$x = 4 \cos \theta, y = 3 \sin \theta$$

Let A be area. $A = 4 (4 \cos \theta) (3 \sin \theta) = 24 \sin 2\theta$



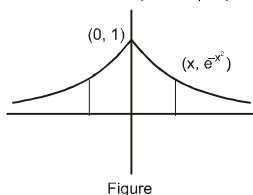
A is maximum when $2\theta = \frac{\pi}{2} \Rightarrow$ Dimensions are. $\frac{2}{\sqrt{2}} \cdot 4, \frac{2.3}{\sqrt{2}}$





E-5. Let A be area $A = (2x)(e^{-x^2})$, $x > 0$

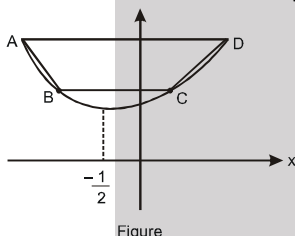
$$\frac{dA}{dx} = -2 \left(x + \frac{1}{\sqrt{2}} \right) \left(x - \frac{1}{\sqrt{2}} \right) e^{-x^2}$$



At $x = \frac{1}{\sqrt{2}}$, A is maximum. Largest area is $2 \frac{1}{\sqrt{2}} e^{-1/2}$

E-6. $f(1^-) \leq f(1)$ and $f(1^+) \leq f(1) - 2 + \log_2(b^2 - 2) \leq 5$
 $0 < b^2 - 2 \leq 128$ $2 < b^2 \leq 130$

E-7. $A(-2, 3) \Rightarrow 3 = 4a - 2b + c$
 $B(-1, 1) \Rightarrow 1 = a - b + c$
 $D(2, 7) \Rightarrow 7 = 4a + 2b + c$
 $\Rightarrow y = x^2 + x + 1$



$C(h, h^2 + h + 1)$, $-1 < h < 2$

$$\text{Area} = \frac{3}{2} (-h^2 + h + 6)$$

Maximum at $h = \frac{1}{2} \Rightarrow C\left(\frac{1}{2}, \frac{7}{4}\right)$

Section (F)

F-1. $f(x) = x^3 - 6x^2 + ax + b$

$f(x)$ satisfies condition in Rolle's theorem on $[1, 3]$

$$f(1) = f(3) \Rightarrow 1 - 6 + a + b = 27 - 54 + 3a + b$$

$$2a = 22$$

$a = 11$ and $b \in \mathbb{R}$.

F-2. $f'(x) = 0 \Rightarrow x = -2, 3$

$$x = -2 \in (-3, 0)$$

$$\therefore c = -2$$

F-3. For $x \in (0, 2)$

$$f'(c) = \frac{f(x) - f(0)}{x - 0}$$

(Here $c \in (0, x)$)

$$\Rightarrow f(x) = 2 \cdot f'(c)$$

$$f(x) \leq 1$$





F-4. Let $f(x) = \frac{ax^5}{5} + \frac{bx^3}{3} + cx$ then $f(1) = 0$, $f(-1) = 0$, $f(0) = 0$

$\Rightarrow$ there exist atleast one root of equation $ax^4 + bx^2 + c = 0$ in $(-1, 0)$ and there exist atleast one root of equation $f'(x) = 0$ in $(0, 1)$

$\Rightarrow$ there exist atleast one root of equation $ax^4 + bx^2 + c = 0$ in $(-1, 0)$ and there exist atleast one root of equation $f'(x) = 0$ in $(0, 1)$

But $\frac{-b}{a}$ equal to negative so equation $ax^4 + bx^2 + c = 0$ has two real and two non-real roots. Hence

there exist exactly one root of equation $ax^4 + bx^2 + c = 0$ in $(-1, 0)$ and there exist exactly one root of equation $f'(x) = 0$ in $(0, 1)$

F-5. $\frac{x^3 - 2x^2 - 5x + 6}{x-1} = \frac{(x-1)(x^2 - x - 6)}{(x-1)} = x^2 - x - 6 \Rightarrow f(x) = \begin{cases} x^2 - x - 6 & ; x \neq 1 \\ -6 & ; x = 1 \end{cases}$

$$\lim_{x \rightarrow 1^+} f(x) = -6; \lim_{x \rightarrow 1^-} f(x) = -6 \Rightarrow \text{LHL} = \text{RHL} = f(1)$$

$\therefore f(x)$ is continuous

LHD at $x = 1$ is 1

RHD at $x = 1$ is 1

$\therefore f(x)$ is differentiable at $x = 1$

$f(-2) = 0$; $f(3) = 0$ all the conditions of Rolle's are holding

$$f'(x) = 2x - 1 = 0$$

$$\Rightarrow x = \frac{1}{2} \therefore \frac{1}{2} \in [-2, 3]$$

PART - III

1. (A) $y^2 = 4ax \Rightarrow \frac{dy}{dx} = \frac{2a}{y}$

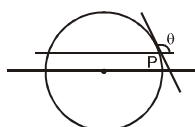
$$y = e^{\frac{-x}{2a}} \Rightarrow \frac{dy}{dx} = \frac{-1}{2a} e^{\frac{-x}{2a}} = \frac{-1}{2a} y$$

$$\text{Product of slopes} = \left(\frac{2a}{y} \right) \left(\frac{-y}{2a} \right) = -1$$

(B) $1 \leq |\sin x| + |\cos x| \leq \sqrt{2}$

$$y = [|\sin x| + |\cos x|] \Rightarrow y = 1$$

P(2, 1)



Figure

$$x^2 + y^2 = 5$$

$$\frac{dy}{dx} = \frac{-x}{y} \tan \theta = \left| \frac{-2-0}{1+0} \right| \operatorname{cosec}^2 \theta = \frac{5}{4}$$



- (C) Let $y^2 = 4a(x + a)$ (1)
 and $y^2 = 4b(x + b)$ (2)
 intersect each other at (h, k) then $h = -(b + a)$
 (h, k) $h = -(b + a)$
 Now $\left. \frac{dy}{dx} \right|_{(h,k)}$ for curve (1) is $\frac{4a}{2k}$ and $\left. \frac{dy}{dx} \right|_{(h,k)}$ for curve (2) is $\frac{4b}{2k}$
 $\Rightarrow \frac{4a}{2k} \times \frac{4b}{2k} = -1 \Rightarrow \frac{4ab}{k^2} = -1 \Rightarrow \frac{4ab}{4a(h+a)} = -1 \Rightarrow \frac{b}{h+a} = -1 \Rightarrow \frac{b}{-b} = -1$
 Which is always true $\Rightarrow \frac{a}{b}$ can take any value from interval, $\mathbb{R} - \{0\}$

- (D) Inverse curves touches each other at line $y = x$
 $\Rightarrow y = x$ is tangent to both curves $\Rightarrow$ equation $x = x^2 + 3x + c$ has both equal roots
 $\Rightarrow c = 1$ and $x = -1 = h$ and $y = -1 = k \Rightarrow |h + k + c| = 1$

2. (A) $f'(x) = 2x - \frac{2}{x^3} = \frac{2(x^2 - 1)(x^2 + 1)}{x^3}$ No point of local maxima

(B) Given expression $= (\sin^{-1}x)^3 + (\cos^{-1}x)^3$
 $= \left(\frac{\pi}{2}\right)^3 - 3\sin^{-1}x \left(\frac{\pi}{2} - \sin^{-1}x\right) \cdot \frac{\pi}{2} = \frac{3\pi}{2} (\sin^{-1}x)^2 - \frac{3\pi^2}{4} \sin^{-1}x + \frac{\pi^3}{8}$

This is quadratic in $\sin^{-1}x$. Therefore it will give maximum value when $\sin^{-1}x = -\frac{\pi}{2} \Rightarrow x = -1$

(C) $f'(x) = 12(x+2)(x+1)(x-1)$

$$\begin{array}{c} + \quad + \quad + \\ - \quad - \quad - \\ \hline -2 \quad -1 \quad 1 \\ \text{sings of } f'(x) \end{array}$$

$\Rightarrow a = -2, b = -1$

(D) $\frac{a^3 + b^3}{48} = \frac{a^3 + (8-a)^3}{48} = \frac{8}{48} (3a^2 - 24a + 64)$

Minimum $\frac{a^3 + b^3}{48}$ is $\frac{8}{48} \frac{(4.3 \cdot 64 - 24^2)}{4.3} = \frac{8}{3}$

3. (A) $f(x)$ is continuous and differentiable $f(0) = f(\pi)$
 Hence condition in Rolle's theorem and LMVT are satisfied.

(B) $f(1^-) = -1, f(1) = 0, f(1^+) = 1$

$f(x)$ is not continuous at $x = 1$, belonging to $\left[\frac{1}{2}, \frac{3}{2}\right]$

Hence, atleast one condition in LMVT and Rolle's theorem is not satisfied

(C) $f'(x) = \frac{2}{5} (x-1)^{-3/5}, x \neq 1$

At $x = 1$, $f(x)$ is not differentiable.

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

(D) At $x = 0$

$$\text{L.H.D.} = \lim_{x \rightarrow 0^-} \frac{x \left(\frac{e^x - 1}{e^x + 1} \right)}{x - 0} = \frac{0 - 1}{0 + 1} = -1$$

R.H.D. = 1

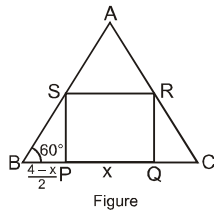
At $x = 0$, $f(x)$ is not differentiable

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.



4. (A) Let $PQ = x$

Then $BP = \frac{4-x}{2}$



$$\therefore PS = \frac{4-x}{2} \tan 60^\circ = \frac{\sqrt{3}}{2} (4-x)$$

$$\therefore \text{area } A \text{ of rectangle} = \frac{\sqrt{3}}{2} (4-x) x$$

$$\frac{dA}{dx} = \frac{\sqrt{3}}{2} (4-2x) = 0 \Rightarrow x = 2 \quad \frac{d^2A}{dx^2} = -\sqrt{3} < 0$$

$\therefore A$ is maximum, when $x = 2$.

$$\therefore \text{Maximum area} = \frac{\sqrt{3}}{2} \cdot 2 \cdot 2 = 2\sqrt{3}$$

$$\text{Square of maximum area} = 12$$

(B) Dimensions be $x, 2x, h$

$$72 = x \cdot 2x \cdot h$$

$$36 = x^2 h \quad \dots (1)$$

$$S = 4x^2 + 6xh$$

$$S = 4x^2 + 6 \frac{36}{x}$$

$$\frac{dS}{dx} = 8x - \frac{216}{x^2} = \frac{8(x^3 - 3^3)}{x^2}$$

For least S , $x = 3$ and least S is 108.

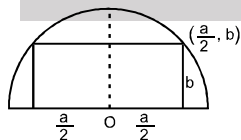
(C) $f(x) = x^3 y = x^3(60-x)$

$$f'(x) = 4x^2(45-x)$$

$f(x)$ is maximum at $x = 45$

(D) $x^2 + y^2 = 5$

$$\frac{a}{2} = \sqrt{5} \cos \theta, b = \sqrt{5} \sin \theta$$



Let $f(\theta)$ be perimeter $f(\theta) = 2a + 2b = 2\sqrt{5} (2\cos\theta + \sin\theta)$

$$f'(\theta) = 2\sqrt{5} (-2\sin\theta + \cos\theta)$$

$$f''(\theta) = 2\sqrt{5} (-2\cos\theta - \sin\theta)$$

$$f'(\theta) = 0 \Rightarrow \tan \theta = \frac{1}{2} \quad \text{and} \quad f''(\theta) < 0 \Rightarrow f(\theta) \text{ is greatest}$$

$$a = 4, b = 1$$

$$a^3 + b^3 = 65$$



EXERCISE # 2

1. $\frac{y}{b} = 1 - \frac{x}{a} \quad \frac{y}{b} = e^{-x/a} \Rightarrow e^{-x/a} = 1 - \frac{x}{a}$ put $t = -\frac{x}{a}$ $e^t = 1 + t$

Draw graph of $y = e^t$, $y = 1 + t$

From graph it is clear that $t = 0$ is the only Solution

$\Rightarrow x = 0 \quad \Rightarrow y = b \quad (0, b)$

2. $x^3 + y^3 = 8xy$, $y^2 = 4x \Rightarrow x^3 + 8x^{3/2} = 8x \cdot 2x^{1/2}$
 $x^3 - 8x^{3/2} = 0 \Rightarrow x^6 - 64x^3 = 0$
 $\Rightarrow x^3 = 0 \text{ or } x^3 = 64$

$x = 4$, $y = 4$. Point of intersection $(4, 4)$ $3x^2 + 3y^2 y' = 8y + 8xy'$

$y' = -1$

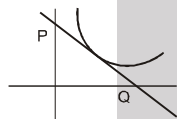
slop of normal is 1

Equation of normal is $y - 4 = 1(x - 4) \Rightarrow x = y$

3. Here $x^{2/3} + y^{2/3} = a^{2/3}$
 Differentiating w.r.t. to x $\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3} \frac{dy}{dx} = 0$

$\frac{dy}{dx} \Big|_{(h, k)} = -\left(\frac{k}{h}\right)^{1/3}$

Equation $(y - k) = -\left(\frac{k}{h}\right)^{1/3} (x - h)$



$P(0, k^{1/3}(h^{2/3} + k^{2/3}))$ or $P(0, a^{2/3} k^{1/3})$

And, $Q(h^{1/3} a^{2/3}, 0)$

$PQ = \sqrt{h^{2/3} a^{4/3} + a^{4/3} k^{2/3}} = |a| = \text{constant.}$

4. The tangent at $(x_1, \sin x_1)$ is $y - \sin x_1 = \cos x_1 (x - x_1)$
 It passes through the origin $\sin x_1 = x_1 \cos x_1 = x_1 \sqrt{1 - \sin^2 x_1}$
 $y_1^2 = \sin^2 x_1 = x_1^2 (1 - y_1^2) \Rightarrow (x_1 y_1)$ lies on the curve
 $y^2 = x^2 (1 - y^2) \Rightarrow x^2 - y^2 = x^2 y^2$

5. $y = e^{[x]} = e^{x-a}$ in $x \in [a, a+1)$ $\frac{dy}{dx} = e^{x-a} = e^{[x]}$

equation of tangent $(Y - y) = \frac{dy}{dx} (X - x)$

passing through $(-1/2, 0)$ $(0 - y) = e^{[x]} (-1/2 - x)$

$\Rightarrow -1 = -\frac{1}{2} - x \Rightarrow x = \frac{1}{2}$

$\therefore$ point $\left(\frac{1}{2}, e^{1/2}\right)$ Number of tangent = 1



6. Let $y = mx + c$ be tangent touching both branches.

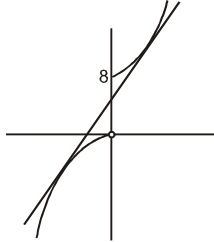
$$f(x) = -x^2, y = mx + c, x < 0$$

$$x^2 + mx + c = 0, \quad m > 0 \quad (\because x < 0) \text{ (negative roots)}$$

$$D = 0 \Rightarrow m^2 = 4c$$

$$f(x) = x^2 + 8, y = mx + c, x > 0$$

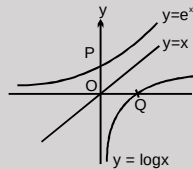
$$x^2 - mx + 8 - c = 0, \quad m > 0 \quad \text{(positive roots)}$$



Figure

$$D = 0 \Rightarrow m^2 = 32 - 4c \Rightarrow c = 4, m^2 = 16 \Rightarrow c = 4, m = 4$$

7. $f(x) = e^x$ & $g(x) = \ln x$ are image of each other in line mirror $y = x$ hence minimum distance between these will be equal to distance between parallel tangents of $f(x)$ & $g(x)$ which are parallel $y = x$.

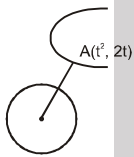


$$\Rightarrow e^x = 1 \text{ \& } = 1$$

$$\Rightarrow x = 0 \text{ \& } x = 1$$

$$P = (0, 1) ; Q = (1, 0); PQ = \sqrt{2}$$

8.



shortest distance always lie along the common normal

Equation of normal at $(t^2, 2t)$ to the parabola is $y + xt = 2t + t^3$ (i)

above equation passes through the center of the circle $c(0, 12)$

$$\therefore 12 = 2t + t^3$$

$$t^3 + 2t - 12 = 0$$

$$t = 2$$

$$\therefore \text{point is } (4, 4)$$

9. $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{\lfloor a^{|x|} \operatorname{sgn} x \rfloor}$

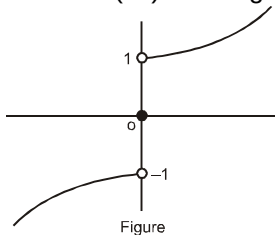
for $a > 1$, $a \neq 1$ and $x \in \mathbb{R}$

$$\ln a \cdot h(x) = \ln f(x) + \ln g(x)$$

$$\Rightarrow (\ln a) h(x) = \{ a^{|x|} \operatorname{sgn} x \} \ln a + \lfloor a^{|x|} \operatorname{sgn} x \rfloor \ln a \Rightarrow h(x) = \{ a^{|x|} \operatorname{sgn} x \} + \lfloor a^{|x|} \operatorname{sgn} x \rfloor$$

$$\Rightarrow h(x) = a^{|x|} \operatorname{sgn} x$$

Now $h(-x) = a^{|-x|} \operatorname{sgn} (-x) = -h(x) \Rightarrow h(x)$ is an odd function Also graph of $h(x)$ is



Figure

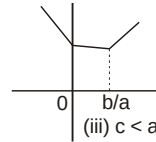
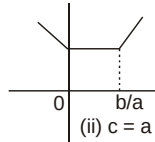
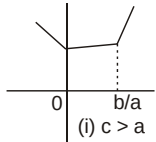
It is clear from the graph that $h(x)$ is an increasing function

10. $x > 1 \Rightarrow f(x) \geq f(1)$



$$\begin{aligned}
 x > 1 &\Rightarrow g(x) \leq g(1) \Rightarrow f(g(x)) \leq f(g(1)) \\
 &\Rightarrow h(x) \leq 1 \quad \dots (i) \\
 \text{Range of } h(x) \text{ is subset of } [1, 10] &\Rightarrow h(x) \geq 1 \quad \dots (ii) \\
 \text{By (i), (ii) we have } h(x) &= 1 \Rightarrow h(2) = 1
 \end{aligned}$$

$$11. \quad f(x) = \begin{cases} b - (a+c)x, & x < 0 \\ b + (c-a)x, & 0 \leq x < b/a \\ (a+c)x - b, & x \geq b/a \end{cases}$$



$$12. \quad f'(x) = \frac{e^x + e^{-x}}{2} > 0$$

$\therefore f(x)$ increasing hence $g(x)$ is also increasing function

$$13. \quad f'(x) = 3x^2 - 3p^2x + 3p^2 - 3 = 3((x-p)^2 - 1) = 3(x-(p+1))(x-(p-1)) \\
 \Rightarrow p-1 > -2 \text{ and } p+1 < 4 \Rightarrow p > -1 \text{ and } p < 3 \Rightarrow -1 < p < 3$$

$$14. \quad f'(x) = \frac{x^{1/x}}{x^2} (1 - \ln x) \quad f'(x) \leq 0, \text{ when } x \geq e$$

$\therefore f(x)$ is decreasing function, when $x \geq e$

$$\pi > e \Rightarrow f(\pi) < f(e)$$

$$\pi^{1/\pi} < e^{1/e} \Rightarrow e^\pi > \pi^e$$

$\therefore$ Statement-1 is True, Statement-2 is False

$$15. \quad \frac{x^2 + x + 2}{x^2 + 5x + 6} < 0 \Rightarrow x \in (-3, -2)$$

For maximum or minimum of the function, put $f'(x) = 0 \Rightarrow a^2 - 3x^2 = 0 \Rightarrow x = -\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}$

If $a > 0$, then point of minima is $x = -\frac{a}{\sqrt{3}} \Rightarrow -3 < -\frac{a}{\sqrt{3}} < -2$ or $2\sqrt{3} < a < 3\sqrt{3}$

if $a < 0$, then point of minima is $x = \frac{a}{\sqrt{3}} \Rightarrow -3 < \frac{a}{\sqrt{3}} < -2 \Rightarrow -3\sqrt{3} < a < -2\sqrt{3}$

$$16. \quad f'(x) = \sin x \cos x (3 \sin x + 2\lambda) \\ f'(x) = 0$$

$$\Rightarrow \sin x = 0 \quad \text{or} \quad \cos x = 0 \quad \text{or} \quad \sin x = \frac{-2\lambda}{3}$$

$$\Rightarrow x = 0 \quad \text{or} \quad \sin x = \frac{-2\lambda}{3} \quad (\text{as } \cos x = 0 \text{ is not possible}).$$

If $\lambda = 0$ then $f'(x) \geq 0$

$\Rightarrow$ no extrema,
hence $\lambda \neq 0$

$$\Rightarrow -1 < \frac{-2\lambda}{3} < 0 \quad \text{or} \quad 0 < \frac{-2\lambda}{3} < 1$$

$$\Rightarrow 0 < \lambda < \frac{3}{2} \quad \text{or} \quad -\frac{3}{2} < \lambda < 0$$

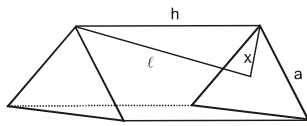
$$17. \quad \ell^2 = h^2 + x^2$$





Area of base (triangle) is $\frac{\sqrt{3}}{4} a^2$

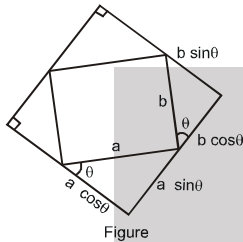
$$3x = \frac{\sqrt{3}}{2} a \quad \text{Volume } V = \frac{\sqrt{3}}{4} h a^2 = h \frac{\sqrt{3}}{4} \cdot 4 \cdot 3x^2 = 3\sqrt{3} h (\ell^2 - h^2) \frac{dV}{dh} = 3\sqrt{3} (\ell^2 - 3h^2)$$



Figure

V is maximum when $h = \frac{\ell}{\sqrt{3}}$.

18.

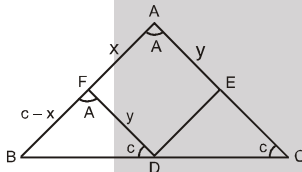


Figure

$$\text{Area} = ab + \left(\frac{1}{2} a^2 \sin \theta \cos \theta + \frac{1}{2} b^2 \sin \theta \cos \theta \right) \cdot 2 = ab + \frac{(a^2 + b^2)}{2} \sin 2\theta$$

$$\text{Maximum area is } ab + \frac{(a^2 + b^2)}{2}$$

19.



Figure

From similar triangles $\triangle ABC, \triangle FBD$ $\frac{c-x}{y} = \frac{c}{b}$

$$\text{Area of AFDE} = xy \sin A = \frac{b}{c} (c-x) \sin A$$

It is maximum when $x = \frac{c}{2}$

$$\therefore \text{Maximum area} = \frac{bc}{4} \sin A$$

statement -1 is true

statement-2 is obvious.

20. Let $f(x) = \sqrt{x}$ If $x \in 7$

$$\frac{f(x+1) - f(x)}{x+1-x} = f'(c)$$

$$\sqrt{x+1} - \sqrt{x} = \frac{1}{2\sqrt{c}}, \quad N^2 < x < c < x+1; \quad c > N^2$$

$$\sqrt{x+1} - \sqrt{x} = \frac{1}{2\sqrt{c}} < \frac{1}{2N}$$

21. $f(x) = \frac{\log x}{x}$ is differentiable and continuous for every $x > 0$. Now for RMV to be applicable $f(a) = f(b)$



$$\Rightarrow \frac{\ln a}{a} = \frac{\ln b}{b} \Rightarrow a^b = b^a \Rightarrow a = 2, b = 4$$

hence $a^2 + b^2 = 20$.

22. Let $g(x) = f(x) - x^2 \Rightarrow g(x)$ has atleast 3 real roots which are $x = 1, 2, 3$

By Lagrange mean value theorem (LMVT)

$\Rightarrow g'(x)$ has atleast 2 real roots in $x \in (1, 3) \Rightarrow g''(x)$ has atleast 1 real root in $x \in (1, 3)$

$\Rightarrow f''(x) - 2 = 0$ for atleast 1 real root in $x \in (1, 3) \Rightarrow f''(x) = 2$ for atleast one $x \in (1, 3)$

PART-II

1. Parametric form of curve is $x = 3t^2, y = 2t^3 \Rightarrow \frac{dy}{dx} = t$

Let $P(3t_1^2, 2t_1^3), Q(3t_2^2, 2t_2^3)$

Conditions are

(i) $\left(\frac{dy}{dx} \right)_P \left(\frac{dy}{dx} \right)_Q = -1$

(ii) $\left. \frac{dy}{dx} \right|_P = \text{Slope of line segment PQ}$

$t_1 t_2 = -1 \dots (i);$

$t_1 = \frac{2}{3} \frac{t_2^2 + t_1 t_2 + t_1^2}{t_2 + t_1} \dots (ii)$

$$\Rightarrow 3(-1 + t_1^2) = 2(t_2^2 - 1 + t_1^2)$$

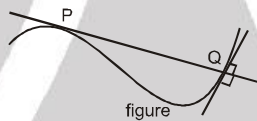
$$t_1^2 = \frac{2}{t_1^2} + 1$$

$$t_1^2 = 2, -1$$

$$t_1^2 = 2 \Rightarrow t_1 = \pm \sqrt{2} \Rightarrow t_2 = \mp \frac{1}{\sqrt{2}}$$

If $t_1 = \sqrt{2}, t_2 = -\frac{1}{\sqrt{2}} \Rightarrow P(6, 4\sqrt{2}), Q\left(\frac{3}{2}, -\frac{1}{\sqrt{2}}\right) \Rightarrow \text{Required line is } y = \sqrt{2}x - 2\sqrt{2}$

If $t_1 = -\sqrt{2}, t_2 = \frac{1}{\sqrt{2}} \Rightarrow P(6, -4\sqrt{2}), Q\left(\frac{3}{2}, \frac{1}{\sqrt{2}}\right) \Rightarrow \text{Required line is } y = -\sqrt{2}x + 2\sqrt{2}$



2. Any point on $y = x^2 + 4x + 8$ is $P(h, h^2 + 4h + 8)$. $\left. \frac{dy}{dx} \right|_P = 2h + 4$

Tangent at P is $(2h + 4)x - y = h^2 - 8$.

It is sufficient to have only one solution for equations $y = x^2 + 8x + 4$,

$$y = (2h + 4)x + 8 - h^2$$

$$\Rightarrow x^2 + (4 - 2h)x + h^2 - 4 = 0 \Rightarrow D = 0$$

$$(4 - 2h)^2 - 4(h^2 - 4) = 0 \Rightarrow h = 2$$

$$8x - y + 4 = 0$$

coordinates of point of contact are $(2, 20)$ and $(0, 4)$

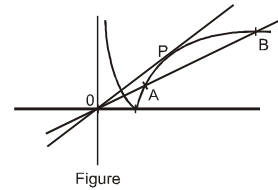
3. $|\ln x| = px$

It is sufficient to find values of p for which $y = |\ln x|$ and $y = px$ has three points in common.



If $y = px$ is passing through points O, A, B then we obtain three roots. Let us consider line $y = px$. When it is passing through points O, P. (tangent)

Let $P(\alpha, p\alpha) (\alpha > 1) \Rightarrow p\alpha = |\ln \alpha|$



$$p\alpha = \ln \alpha \quad \dots(1)$$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$p = \frac{1}{\alpha} \quad \dots(2)$$

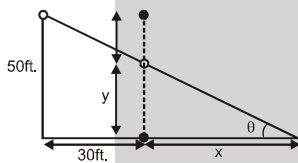
$$(1), (2) \Rightarrow \alpha = e$$

Slope of tangent (OP) is $\frac{1}{e}$.

For three roots condition is $0 < p < \frac{1}{e}$.

4. $y = 50 - 16t^2$ So, $\frac{dy}{dt} = -32t$

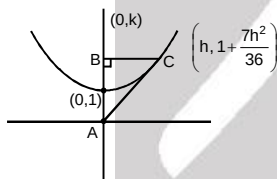
$$\tan \theta = \frac{y}{x} = \frac{50}{30+x} \Rightarrow y = \left(\frac{50}{30+x} \right) \cdot x$$



Figure

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{1500}{(30+x)^2} \cdot \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = -16 \frac{(375)^2}{1500} = -1500 \text{ ft/sec} = 100\lambda \text{ ft/sec} \Rightarrow \lambda = -15 \Rightarrow |\lambda| = 15$$

5. at $t = 0$; $x = 0, y = 1$



$$\frac{dy}{dt} = 2 \text{ cm/sec}$$

$$A = \frac{1}{2} \times h \times \left(1 + \frac{7h^2}{36} \right)$$

$$\frac{dA}{dt} = \left(\frac{1}{2} \left(1 + \frac{7h^2}{36} \right) + \frac{h}{2} \left(\frac{14h}{36} \right) \right) \frac{dh}{dt}; \frac{dA}{dt} = \left(\frac{1}{2} \times 8 + 3 \times \frac{14 \times 6}{36} \right) \times \frac{6}{(7/2)}$$

(At, $t = 7/2$ sec, change in y-co-ordinate = 7 hence, pt. C has

$$y\text{-co-ordinate} = 8 \text{ and } x\text{-co-ordinate} = 6 \text{ at } t = 7/2 \text{ sec.}) = (4 + 7) \times \frac{6}{7} \times 2 = \frac{132}{7} \text{ cm}^2/\text{sec}$$



$$6. \quad f'(x) = \frac{2x(e^{x^2} + e^{-x^2}) (e^{x^2} + e^{-x^2}) - (e^{x^2} - e^{-x^2}) (2xe^{x^2} - 2xe^{-x^2})}{(e^{x^2} + e^{-x^2})^2}$$

$$= \frac{2x \left((e^{x^2} + e^{-x^2})^2 - (e^{x^2} - e^{-x^2})^2 \right)}{(e^{x^2} + e^{-x^2})^2} = \frac{8x}{(e^{x^2} + e^{-x^2})^2} \geq 0 \quad x \in [0, \infty) \Rightarrow \text{least value of } \alpha \text{ is } 2.$$

$$7. \quad \text{Let } f(x) = 2 \sin x + \tan x - 3x$$

$$f'(x) = 2 \cos x + \sec^2 x - 3 = \frac{(\cos x - 1)^2 (2 \cos x + 1)}{\cos^2 x} > 0$$

$f(x)$ is M.I.

$x > 0$

$f(x) > f(0)$

$2 \sin x + \tan x > 3x$

$$3x < 2 \sin x + \tan x \Rightarrow \frac{3x}{2 \sin x + \tan x} < 1 \text{ for } x \in \left(0, \frac{\pi}{2}\right)$$

$$\text{and } \lim_{x \rightarrow 0^+} \frac{3x}{3 \sin x + \tan x} = 1 \Rightarrow \lim_{x \rightarrow 0^+} \left[\frac{3x}{3 \sin x + \tan x} \right] = 0$$

$$\text{and } \lim_{x \rightarrow 0^+} \frac{\tan^3 x - \sin^3 x}{x^5} = \lim_{x \rightarrow 0^+} \frac{\tan^3 x (1 - \cos^3 x)}{x^5} = \lim_{x \rightarrow 0^+} \frac{\tan^3 x}{x^3} \cdot \frac{(1 - \cos x)}{x^2} (\cos^2 x + \cos x + 1) = \frac{3}{2}$$

$$\text{Hence } \lim_{x \rightarrow 0^+} \left(2 + \left[\frac{3x}{2 \sin x + \tan x} \right] \right)^{\frac{\tan^3 x - \sin^3 x}{x^5}} = (2 + 0)^{3/2} = 2\sqrt{2} = 2.828$$

$$8. \quad f(x) = 2e^x - ae^{-x} + (2a - 3)x - 3$$

$$f'(x) = 2e^x + ae^{-x} + (2a - 3) \geq 0 \quad \forall x \in \mathbb{R} \Rightarrow a \geq \frac{e^x(3 - 2e^x)}{(1 + e^x)}$$

$$\text{Let } y = \frac{e^x(3 - 2e^x)}{(1 + e^x)} \quad \text{Let } e^x = t$$

$$y = t \left(\frac{3 - 2t}{1 + t} \right) \quad t \in (0, \infty) \quad \frac{dy}{dt} = -\frac{(2t + 3)(2t - 1)}{(1 + t)^2}$$

$$\text{hence maximum value of } y \text{ will be at } t = \frac{1}{2} \quad y_{\max} = \frac{1}{2} \quad \text{hence minimum value of } a \text{ is } \frac{1}{2}$$

$$9. \quad f'(x) = 4 [\cos^2 x - 2(a + 1)\cos x + a^2 + 2a - 4]$$

$$\text{Let } \phi(t) = t^2 - 2(a + 1)t + a^2 + 2a - 4, \quad -1 \leq t \leq 1$$

It is sufficient to find values of t when $\phi(t) = 0$ has no root in $[-1, 1]$

$$D = 20$$

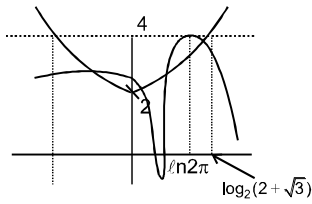
Case-I : $\phi(-1) > 0$, $a + 1 < -1$, $D = 20 > 0$

$$\Rightarrow a \in (-\infty, -2 - \sqrt{5}) \cup (-2, +\sqrt{5}, \infty) \text{ and } a < -2 \Rightarrow a \in (-\infty, -2 - \sqrt{5})$$

Case - II : $\phi(1) > 0$, $a + 1 > 1$, $D = 20 > 0 \Rightarrow a \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty)$ and $a > 0 \Rightarrow a \in (\sqrt{5}, \infty)$



10. Using graph of expressions on both the sides, we get only two roots.



11. $-1 \leq p \leq 1$
Consider $f(x) = 4x^3 - 3x - p = 0$
 $f\left(\frac{1}{2}\right) \leq 0$
 $f(1) \geq 0$

$\therefore f(x)$ has at least one root between $\left[\frac{1}{2}, 1\right]$

Also $f'(x) = 12x^2 - 3 > 0 \forall \left[\frac{1}{2}, 1\right] \Rightarrow f$ is increasing on $\left[\frac{1}{2}, 1\right]$

$\Rightarrow f(x)$ has only one real root between $\left[\frac{1}{2}, 1\right]$ To find root put $x = \cos \theta$

$\Rightarrow \cos 3\theta = p \Rightarrow \theta = \frac{1}{3} \cos^{-1} p$

$\therefore$ Root is $\cos\left(\frac{1}{3} \cos^{-1} p\right)$

12. Let $y = 2^{x^2} - 1 + \frac{200}{2^{x^2} + 1}$

let $2^{x^2} = t \Rightarrow y = t - 1 + \frac{200}{t + 1} \quad t \in [1, \infty) \Rightarrow \frac{dy}{dt} = \frac{(t+1)^2 - 200}{(t+1)^2} = \frac{(t+1+10\sqrt{2})(t+1-10\sqrt{2})}{(t+1)^2}$

$\Rightarrow$ hence function has minimum value at $t = -1 + 10\sqrt{2} \quad y_{\min.} = 20\sqrt{2} - 2 = 26.28$

13. $f(x) = (x-1)^{2013} + (x-2)^{2013} + \dots + (x-2013)^{2013}$
 $f'(x) = 2013 [(x-1)^{2012} + (x-2)^{2012} + \dots + (x-2013)^{2012}]$
 $f'(x) > 0 \Rightarrow f(x) \uparrow$

$f(x) \in (-\infty, \infty) \Rightarrow f(x)$ can have only one real root

Due to symmetric nature, real root is $\frac{1+2013}{2} = \frac{2014}{2} = 1007$

14. $f(x) = \frac{a}{3} x^3 + (a+2)x^2 + (3a-10)x$

$f'(x) = g(x) = ax^2 + 2(a+2)x + (a-1) = a(x-\alpha)(x-\beta) \quad (\beta > \alpha)$

Case - I : If $a = 0 \Rightarrow f'(x) = 4x - 10 \Rightarrow x = \frac{10}{4}$ is point of minimum (which is not negative)

Case- II : If $a > 0$ then β will be point of minima which is negative hence both root of $g(x) = 0$

must be negative. $\left. \begin{array}{l} g(0) > 0 \\ D > 0 \\ -b/2a < 0 \end{array} \right\} \Rightarrow a \in \left(\frac{10}{3}, \frac{7+\sqrt{57}}{2}\right)$

Case-III : If $a < 0$ then α will be point of minima which is negative hence β can be negative, zero or positive but $g(0) < 0$ is hence β can be negative only hence again $g(x) = 0$ must have both negative

roots $\left. \begin{array}{l} g(0) < 0 \\ D > 0 \\ -b/2a < 0 \end{array} \right\} \Rightarrow a \in \phi$ hence exhaustive set of values of a is $\left(\frac{10}{3}, \frac{7+\sqrt{57}}{2}\right)$



15. $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$

$$\lim_{x \rightarrow 0} \left(1 + \frac{f(x)}{x^3} \right) = 1 \Rightarrow a_0 = a_1 = a_2 = a_3 = 0$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{f(x)}{x^3} \right)^{1/x} = e^2$$

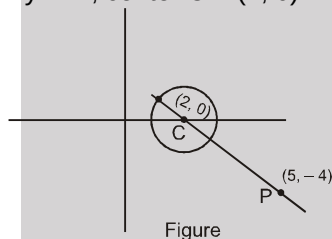
$$\lim_{x \rightarrow 0} e^{(a_4 + a_5x + a_6x^2)} = e^2 \Rightarrow a_4 = 2$$

$$f(x) = 2x^4 + a_5x^5 + a_6x^6$$

$$f'(x) = x^3(8 + 5a_5x + 6a_6x^2) \quad f'(1) = 0, \quad f'(2) = 0$$

$$a_5 = -\frac{12}{5}, \quad a_6 = \frac{2}{3} \quad f(x) = 2x^4 - \frac{12}{5}x^5 + \frac{2}{3}x^6$$

16. Let $y = \sqrt{-3 + 4x - x^2}$
 $x^2 + y^2 - 4x + 3 = 0$
 $(x - 2)^2 + y^2 = 1$, center $C = (2, 0)$



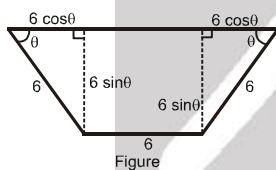
Consider point $P(5, -4)$

$$CP = \sqrt{9 + 16} = 5$$

$$\text{Maximum value of } \left(\sqrt{-3 + 4x - x^2} + 4 \right)^2 + (x - 5)^2 \text{ is } (5 + 1)^2 = 36.$$

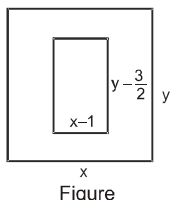
17. Area = $36(1 + \cos\theta) \sin\theta = 36 \cdot 4 \cdot \cos^3(\theta/2) \sin(\theta/2)$

$$\text{Maximum occurs when } \tan\left(\frac{\theta}{2}\right) = \sqrt{\frac{1}{3}} \Rightarrow \theta = \frac{\pi}{3}$$



18. $xy = 18$

$$\text{Area of printed space} = (x - 1) \left(y - \frac{3}{2} \right) = 18 + \frac{3}{2} - \left(\frac{3x}{2} + \frac{18}{x} \right)$$



$$\text{Maximum when } \frac{3x}{2} = \frac{18}{x} \Rightarrow x = 2\sqrt{3} \quad y = 3\sqrt{3}$$



19. Fuel charges per hour = $kv^2 \Rightarrow 48 = k \cdot 16^2$

$\Rightarrow$ Fuel charges per hour = $\frac{3}{16} v^2$

Charges per hour = $\frac{3}{16} v^2 + 300$

Expenses of journey = $(\frac{3}{16} v^2 + 300) \frac{s}{v}$

where v = speed s = distance

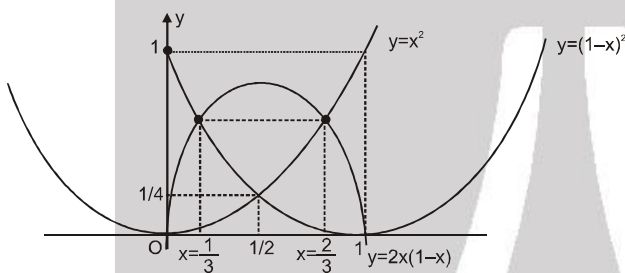
Maximum occurs when $\frac{3v}{16} = \frac{300}{v}$

($\because ax + \frac{b}{x}$, $a, b, > 0, x > 0$, has minimum when $ax = \frac{b}{x}$)

$v^2 = 16 \cdot 100$

$v = 40$

20.



From figure we can see Rolle's theorem is applicable for $x \in [\frac{1}{3}, \frac{2}{3}]$ and $f'(c) = 0 = 2 - 4c$

$\Rightarrow c = \frac{1}{2} \quad a + b + c = \frac{1}{3} + \frac{2}{3} + \frac{1}{2} = \frac{3}{2}$

21. There exist atleast one $c \in (\alpha, \alpha + d)$ such that $f'(c) = \frac{f(\alpha + d) - f(\alpha)}{d} \Rightarrow f'(c) \in \left[\frac{-6}{d}, \frac{6}{d} \right]$.

If d is very large then $f'(c)$ will approach to zero. Also $(f'(\alpha))^2 \in [77, 83]$ Because $f'(x)$ is continuous function so $(f'(x))^2$ can take all values from $(0, 77] \Rightarrow (f'(x))^2$ can take 76 integral values in $(0, 77)$

PART - III

1. $2y^3 = ax^2 + x^3 \Rightarrow 6y^2 \frac{dy}{dx} = 2ax + 3x^2 \Rightarrow \frac{dy}{dx} \Big|_{(a, a)} = \frac{5a^2}{6a^2} = \frac{5}{6}$

Tangent at (a, a) is $5x - 6y = -a \Rightarrow \alpha = \frac{-a}{5}, \beta = \frac{a}{6}$

$\alpha^2 + \beta^2 = 61 \Rightarrow \frac{a^2}{25} + \frac{a^2}{36} = 61$
 $a^2 = 25.36$
 $a = \pm 30$

2. $x = 2 \Rightarrow t^2 + 3t - 10 = 0 \Rightarrow t = 2, -5$
 $y = -1 \Rightarrow t^2 - t - 2 = 0 \Rightarrow t = 2, -1$
 $\Rightarrow t = 2$ (common value)

$\frac{dy}{dx} = \frac{4t-2}{2t+3} \Rightarrow \frac{dy}{dx} \Big|_{t=2} = \frac{6}{7} = \frac{-1}{6} = -\frac{7}{6} = \frac{-1}{6} \sqrt{1 + \frac{36}{49}} = -\frac{\sqrt{85}}{6}$



3. Let $f(x) = x + \sin x \Rightarrow f'(x) = 1 + \cos x$
 As $f'(x) \geq 0 \forall x \in \mathbb{R}$, $f(x)$ is increasing
 Let $g(x) = \sec x$
 $g'(x) = \sec x \tan x$
 $g'(x)$ changes sign.
 $g(x)$ is neither increasing nor decreasing.

4. $f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{x^2+1}} = 1 - \frac{1}{1+x^2} + 1 - \frac{1}{\sqrt{x^2+1}} = \frac{x^2}{1+x^2} + \left(1 - \frac{1}{\sqrt{x^2+1}}\right) \geq 0$

5. $g(x) = 2f\left(\frac{x}{2}\right) + f(1-x)$ and $g'(x) = f'(x/2) - f'(1-x)$

Now $g(x)$ is increasing if $g'(x) \geq 0$

$$f'\left(\frac{x}{2}\right) \geq f'(1-x)$$

[$\because f''(x) < 0$ i.e. $f'(x)$ is decreasing]

$$\Rightarrow \frac{x}{2} \leq 1-x \Rightarrow x \leq 2-2x \Rightarrow 3x \leq 2 \Rightarrow x \leq 2/3 \Rightarrow 0 \leq x \leq \frac{2}{3} \Rightarrow g(x) \text{ increases in } 0 \leq x \leq 2/3$$

$$\text{and } g'(x) \leq 0 \text{ for decreasing} \Rightarrow f'\left(\frac{x}{2}\right) \leq f'(1-x) \Rightarrow \frac{x}{2} \geq 1-x \Rightarrow x \geq 2/3 \Rightarrow 2/3 \leq x \leq 1$$

6. $f'(x) = \frac{m}{n} x^{\left(\frac{m-n}{n}\right)}$

$m-n$ is odd.

$$f'(x) < 0 \quad \forall x \in (-\infty, 0)$$

$$f'(x) > 0 \quad \forall x \in (0, \infty)$$

7. Let $h(x) = f(x) g(x)$

$$h'(x) = f'(x) g(x) + g'(x) f(x)$$

$$\text{As } f'(x) < 0, g(x) \leq 0 \Rightarrow f'(x) g(x) \geq 0 \text{ and } g'(x) > 0, f(x) \geq 0 \Rightarrow f(x) g'(x) \geq 0$$

$$\Rightarrow h'(x) \geq 0 \Rightarrow h(x) \text{ is increasing.}$$

$$\text{Let } x_1, x_2 \in I$$

$$x_1 < x_2$$

$$g(x_1) < g(x_2)$$

$$f(g(x_1)) > f(g(x_2))$$

$$f(g(x_1)) > f(g(x_2))$$

$$\Rightarrow f(g(x)) \text{ is monotonically decreasing.}$$

8. $\phi'(x) = (3(f(x))^2 - 6(f(x)) + 4)f'(x) + 5 + 3 \cos x - 4 \sin x$

$$5 - \sqrt{9+16} \leq 5 + 3 \cos x - 4 \sin x \leq 5 + \sqrt{9+16}$$

$$\text{adding } (3(f(x))^2 - 6(f(x)) + 4)f'(x)$$

$$(3(f(x))^2 - 6(f(x)) + 4)f'(x) \leq \phi'(x) \leq (3(f(x))^2 - 6(f(x)) + 4)f'(x) + 10$$

$$\because 3(f(x))^2 - 6f(x) + 4 = 3(f(x) - 1)^2 + 1 > 0$$

$$(3(f(x))^2 - 6(f(x)) + 4)f'(x) \geq 0 \quad \text{when ever } f(x) \text{ is increasing.}$$

$$\Rightarrow \phi'(x) \geq 0 \Rightarrow \phi(x) \text{ is increasing, when ever } f(x) \text{ is increasing.}$$

$$\text{If } f'(x) = -11 \text{ then}$$

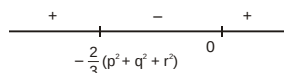
$$(3(f(x))^2 - 6f(x) + 4)f'(x) + 10 = -33(f(x) - 1)^2 - 1 < 0$$

$$\Rightarrow \phi'(x) < 0 \Rightarrow \phi(x) \text{ is decreasing.}$$



$$9. \quad f(x) = \begin{vmatrix} x+p^2 & pq & pr \\ pq & x+q^2 & qr \\ pr & qr & x+r^2 \end{vmatrix} = x^3 + (p^2 + r^2 + q^2) x^2$$

$$f'(x) = 3x^2 + 2x(p^2 + q^2 + r^2) = x \{3x + 2(p^2 + q^2 + r^2)\}$$



Here $f(x)$ is increasing if $x < -\frac{2}{3}(p^2 + q^2 + r^2)$ and $x > 0$

decreasing is if $-(p^2 + q^2 + r^2) < x < 0$

$$10. \quad f(x) = \frac{x^2}{4} \operatorname{cosec}^2 \frac{x}{2}$$

$$f'(x) = \frac{x}{2} \operatorname{cosec}^2 \frac{x}{2} \cot \frac{x}{2} \left(\tan \frac{x}{2} - \frac{x}{2} \right)$$

For $0 < x < 1$, $\tan \frac{x}{2} > \frac{x}{2} \Rightarrow f'(x) > 0 \Rightarrow f(x)$ is increasing.

$$11. \quad (A) \quad y = \frac{2x^2 - 1}{x^4} \text{ is even function.}$$

Even function is nonmonotonic.

[True]

(B) If range of $f(x)$ is not $\mathbb{R}$ and c does not belong to range of $f(x)$ then it is not necessary to have one

[False]

$$(C) \quad f'(x) = 1 > 0$$

$f(x)$ is increasing

$f(0), f(1)$ is not defined. Hence no local maxima/minima.

[True]

$$(D) \quad f(6) = 0; f(6-h) > f(6)$$

$$\& f(6+h) > f(6)$$

Minima at $x = 6$

[False]

$$12. \quad f'(x) = \frac{\sin x - x \cos x}{\sin^2 x}$$

$$\text{Let } g(x) = \sin x - x \cos x$$

$$g'(x) = \cos x + x \sin x - \cos x$$

$$g'(x) > 0 \quad \forall x \in (0, \pi/2)$$

$$g(x) \uparrow$$

$$g(0) < g(x)$$

$$0 < \sin x - x \cos x$$

$$\therefore f'(x) > 0$$

$$f(x) \uparrow \quad \text{Let } \alpha = \pi/12, \beta = \pi/4, r = 5\pi/12$$

$$\Rightarrow f(\alpha) < f(\beta) < f(r)$$

$$\text{Let } H(x) = 2(x - f(\beta))(x - f(r)) + 3(x - f(\alpha))(x - f(r)) + 4(x - f(\alpha))(x - f(\beta))$$

$$H(f(r)) > 0, H(f(\beta)) < 0, H(f(\alpha)) > 0$$

$$\Rightarrow \text{one root in } (f(r), f(\beta)) \quad (f(r), f(\beta))$$

$$\text{and other in } (f(\beta), f(r)) \quad (f(\beta), f(r))$$



13. $f'(x) = (x-1)^{n-1} (x+1)^{n-1} [2(n+1)x^3 + (2n+1)x^2 + 2(n-1)x - 1]$

At $x = 1$ $2(n+1)x^3 + (2n+1)x^2 + 2(n-1)x - 1 \neq 0$

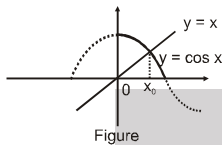
for $n \in \mathbb{N}$

$\therefore n-1$ must be odd

$\Rightarrow n$ is even

14. $f'(x) = \frac{\sec^2 x (\cos x + x) (\cos x - x)}{(1+x \tan x)^2}$

The only factor in $f'(x)$ which changes sign is $\cos x - x$.



Let us consider graph of $y = \cos x$ and $y = x$

It is clear from figure that for $x \in (0, x_0)$, $\cos x - x > 0$ and for $x \in (x_0, \frac{\pi}{2})$

$\cos x - x < 0, \Rightarrow f'(x)$ has maxima at x_0

15. $f'(x) = \frac{a}{x} + 2bx + 1$

$f'(-1) = 0$

$-a - 2b + 1 = 0$

$a + 2b = 1$

$f'(2) = 0$

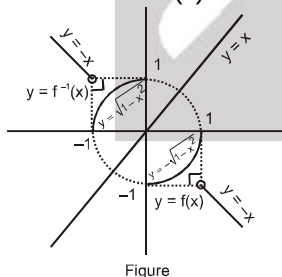
$\frac{a}{x} + 4b + 1 = 0 \Rightarrow a + 8b + 2 = 0$

$-6b = 3 \Rightarrow b = -\frac{1}{2}, a = 2$

16. From graph $f^{-1}(x) = \begin{cases} -x & ; x < -1 \\ \sqrt{1-x^2} & ; -1 \leq x \leq 0 \end{cases}$

Maximum of $f(x)$ exist at $x = 1$

Minimum of $f^{-1}(x)$ exist at $x = -1$



17. $f'(x) = \frac{1}{1+x^2} - \frac{1}{2} \cdot \frac{1}{x}, x > 0 = \frac{-(x-1)^2}{2x(1+x^2)} \leq 0 \quad \forall x > 0.$

$f(x)$ is decreasing $\forall x > 0.$

On $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$, greatest value is $f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} - \frac{1}{2} \ln\left(\frac{1}{\sqrt{3}}\right)$ and least value is $f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln \sqrt{3}$



18. $f'(x) = \frac{-40.12x(x+3)(x-1)}{(3x^4 + 8x^3 - 18x^2 + 60)^2}$

$$f'(x) = 0$$

$$\text{at } x = 0, x = -3, x = 1$$

so at $x = 0$, $f(x)$ has local minima.

and at $x = -3, x = 1$; $f(x)$ has local maxima

$$f(1) = \frac{40}{53}, f(-3) = \frac{-40}{75}, f(-3) < 0, f(1) > 0 \text{ and } f(x) \neq 0$$

$\Rightarrow f(x)$ is undefined at point(s) in $(-3, 1)$. Hence $f(x)$ has no absolute maxima.

19. $f(x) = \frac{x-2}{x+3}; x \neq 1, -3 \Rightarrow f'(x) = \frac{5}{(x+3)^2}$

20. $(2x)^2 + 2.2x \cdot 3y + (3y)^2 + (y-1)(y-3) = 0 \Rightarrow (2x+3y)^2 + (y-1)(y-3) = 0$

$$x \in \mathbb{R}$$

$$\text{So } D \geq 0$$

$$144y^2 - 16(10y^2 - 4y + 3) \geq 0$$

$$16[-y^2 + 4y - 3] \geq 0$$

$$y^2 - 4y + 3 \leq 0$$

$$(y-1)(y-3) \leq 0$$

$$\therefore 1 \leq y \leq 3$$

$$\text{So } y_{\max} = 3 \text{ and } y_{\min} = 1$$

21. $f(0) = 0 \neq f(1)$
there will be no $x \in (0, \infty)$ ($\therefore$ Rolle's theorem is not applicable)

$$\text{for which } f'(x) = 0 \text{ i.e. } \cot^{-1} x = \frac{x}{1+x^2}$$

$$f'(x) = \frac{-1}{1+x^2} - \frac{(1+x^2) - 2x^2}{(1+x^2)^2} = \frac{-1}{1+x^2} + \frac{x^2-1}{(x^2+1)^2}$$

$$f''(x) = \frac{-2}{(x^2+1)^2} < 0$$

$$f'(x) \text{ is strictly decreasing } \lim_{x \rightarrow \infty} f'(x) = \lim_{x \rightarrow \infty} \left(\frac{-x}{1+x^2} + \cot^{-1} x \right) = 0$$

$$f(0+) = \lim_{x \rightarrow 0^+} \left(\cot^{-1} x - \frac{x}{1+x^2} \right) = \frac{\pi}{2} \quad \frac{f\left(x + \frac{2}{\pi}\right) - f(x)}{2/\pi} = f'(c) \quad c \in \left(0, \frac{\pi}{2}\right) \quad (\therefore \text{LMVT is applicable})$$

$$\therefore f'(c) < \frac{\pi}{2}$$

$$f\left(x + \frac{2}{\pi}\right) - f(x) < \frac{2}{\pi} \times \frac{\pi}{2}$$

$$f\left(x + \frac{2}{\pi}\right) - f(x) < 1$$

$$f(x) \geq 0; f(x) \text{ is increasing}$$

$$f(x) \in [f(0), f(\infty))$$

$$f(0) = 0$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x \cot^{-1} x = \lim_{x \rightarrow 0} \frac{\cot^{-1} x}{1/x} = \lim_{x \rightarrow \infty} \frac{-1}{1+x^2} \times (-x^2) = 1$$

$$f(x) \in [0, 1) \Rightarrow f(x) = \sec x \text{ will have no solution}$$



22. $f(4) = f(5) = f(6) = f(7) = 0$

By Rolle's theorem on interval

$[4, 5], [5, 6], [6, 7]$ we have

$f'(x) = 0$ for at least once in each intervals $(4, 5), (5, 6), (6, 7)$.

23. (A) let $f(x) = \tan^{-1}x$

$$f'(x) = \frac{1}{1+x^2}$$

$$\therefore |f'(c)| = \frac{1}{1+c^2} < 1$$

$$\left| \frac{\tan^{-1}x - \tan^{-1}y}{x - y} \right| < 1$$

(B) Let $f(x) = x^{100} + \sin x - 1$

$$f'(x) = 100x^{99} + \cos x > 0, x \in [0, 1]$$

$\Rightarrow f(x)$ is increasing.

(C) Suppose $f(x) = ax^3 - 2bx^2 + cx$, then clearly $f(0) = 0$

$$\text{and } f(1) = a - 2b + c = 0,$$

$$\therefore f(0) = f(1)$$

$$\therefore \text{By Rolle's theorem } f'(x) = 3ax^2 - 4bx + c = 0$$

for atleast one x in $(0, 1)$ which is positive

(D) $f(x) = 3 \tan x + x^3 - 2$

$$\Rightarrow f'(x) = 3 \sec^2 x + 3x^2 > 0$$

$\Rightarrow f(x)$ is always increasing

24. Apply Roll's theorem on $f(x)$, $g(x) = e^x f(x)$ and $h(x) = e^{-x} f(x)$

25. (A) Let $x \Rightarrow x + h$ and $y \rightarrow x$

$$|\tan^{-1}x - \tan^{-1}y| \leq |x - y|$$

$$|\tan^{-1}(x + h) - \tan^{-1}x| \leq |h|$$

$$\left| \frac{d}{dx}(\tan^{-1}x) \right| \leq 1$$

$$\left| \frac{1}{1+x^2} \right| \leq 1 \quad \text{hence true}$$

(C) $|\sin x - \sin y| \leq |x - y|$

$$x \rightarrow x + h \quad y \rightarrow x$$

$$\left| \frac{\sin(x+h) - \sin x}{h} \right| \leq 1$$

$$|\cos x| \leq 1 \quad \text{hence true}$$

Alternative solutions



For $x = y$, this is true

$\therefore$ Let $x, y \in \mathbb{R}$ and $x < y$

consider $f(t) = \tan^{-1}t$, $t \in [x, y]$

Using LMVT, $\frac{\tan^{-1}y - \tan^{-1}x}{y - x} = \frac{1}{1+c^2}$, $c \in (x, y)$

$$\Rightarrow \tan^{-1}y - \tan^{-1}x = \frac{y-x}{1+c^2} \leq y-x \quad \dots\dots(i)$$

similarly $x > y$, $\tan^{-1}x - \tan^{-1}y \leq x-y \quad \dots\dots(ii)$

From (i) and (ii) we get $|\tan^{-1}x - \tan^{-1}y| \leq |x-y|$

Similarly considering $g(t) = \sin t$ in $[x, y]$ we get $\frac{\sin y - \sin x}{y-x} = \cos c$

$$\Rightarrow \sin y - \sin x = (\cos c)(y-x) \leq y-x \quad \dots\dots(iii)$$

$$\text{and } \sin x - \sin y \leq x-y \quad \dots\dots(iv)$$

$$(iii), (iv) \Rightarrow |\sin x - \sin y| \leq |x-y|$$

$$26. \Rightarrow |f'(c)| \leq 2 \Rightarrow |f'(x) - f'(a)| \leq 2|x-a| \Rightarrow |f'(x) - f'(a)| < 2$$

$$a \text{ is that point on the interval when } f'(a) = \left| \frac{f(2) - f(1)}{2-1} \right| = 0 \text{ as } f(1) = f(2)$$

$$27. (A) \text{ Apply L. M. V. T. in } (a, a+6) \text{ we get } f'(c) = \frac{f(a+6) - f(a)}{6}$$

$$\text{Now } f(a+6) - f(a) \in [-6, 6] \Rightarrow f'(c) \in [-1, 1] \Rightarrow (f'(c))^2 \leq 1$$

$$(B) \text{ Apply L. M. V. T. in } (-d, 0) \text{ where } d \text{ is positive then } f'(c) = \frac{f(0) - f(-d)}{d} \Rightarrow f'(c) \in \left[\frac{-6}{d}, \frac{6}{d} \right]$$

If d is approaching to infinity then $f'(c)$ approaches to zero. And $(f'(0))^2 \in [76, 85]$, Also $f'(x)$ is continuous function. So $(f'(x))^2$ can take all real value from $(0, 76]$

$$(C) \text{ By using option (A) we get there exist a value } x_0 \in (-6, 0) \text{ such that } (f'(x))^2 \leq 1$$

$$\text{also } f(x_0)^2 \leq 9 \Rightarrow (f'(x_0))^2 + (f(x_0))^2 \leq 10$$

But $f(x)^2 + (f'(x))^2$ is continuous function and $(f(0))^2 + (f'(0))^2 = 85$ hence there exists $\alpha \in [x_0, 0]$ such that $(f'(\alpha))^2 + (f(\alpha))^2 = 10 \Rightarrow$ there exist $\alpha \in (-6, 0)$ such that $(f'(\alpha))^2 + (f(\alpha))^2 = 10$.

$$(D) \text{ Similarly by using option (C) we say that there exists } \beta \in (0, 6) \text{ such that } (f(\beta))^2 + (f'(\beta))^2 = 10$$

$$\text{Assume } H(x) = (f(x))^2 + (f'(x))^2 \text{ where } H(0) = 85$$

Let p is largest negative number lies between $(-6, 0)$ such that $H(p) = 10$

Similarly let q is smallest positive number lies between $(0, 6)$ such that $H(q) = 10$

Apply Rolle's in $H(x)$ in interval (p, q) we get there exists r in (p, q) such that $H'(r) = 0$

$$\Rightarrow 2f'(r)(f(r) + f''(r)) = 0$$

Because in (p, q) , $(f(x))^2 \leq 9$ & $H(x) \geq 10$

$$\Rightarrow (f'(x))^2 \geq 1 \Rightarrow f'(r) \neq 0 \Rightarrow f(r) + f''(r) = 0 \Rightarrow |f(r)| = |f''(r)|$$

$$(E) \text{ There exist one } c \in (-3, 3) \text{ such that } f'(c) = \frac{f(3) - f(-3)}{6} \in [-1, 1] \text{ and } f'(0)^2 \in [76, 85]$$



let $f'(0) \geq \sqrt{76}$ then $f'(x)$ is mix up of increasing as well as decreasing function

$$\Rightarrow f''(c) < 0 \text{ for some } c \in (-3, 3) \Rightarrow f'(c) f''(c) < 0$$

PART - IV

1. Here, $m = \left. \frac{dy}{dx} \right|_{x=0}$

$$\frac{dy}{dx} = 3x^2 + 6x + 4 \Rightarrow m = 4$$

and, $k = y(0) \Rightarrow k = -1$

$$\ell = |k| \sqrt{1 + \frac{1}{m^2}} \Rightarrow \ell = |(-1)| \sqrt{1 + \frac{1}{16}} = \frac{\sqrt{17}}{4}$$

2. $|yy'| = |y/y'|$ at $(0, 1) \Rightarrow (y') \text{ at } (0, 1) \text{ equal } \pm 1 \Rightarrow (pe^{px} + p)_{(0,1)} = \pm 1 \Rightarrow 2p = \pm 1 \Rightarrow p = \pm 1/2$

3. Length of subnormal = $\left| y \frac{dy}{dx} \right| = \left| -3 \sin\left(\frac{-\pi}{4}\right) \frac{-3 \cos\left(\frac{-\pi}{4}\right)}{\left(-\sqrt{2} \sin\left(\frac{-\pi}{4}\right)\right)} \right| = \frac{3}{\sqrt{2}} \frac{3}{\sqrt{2}} = \frac{9}{2}$

(4 to 6)

Let $g(x) = \frac{x + \sin x}{2}$, $x \in [0, \pi]$. $g(x)$ is increasing function of x .

$\therefore$ range of $g(x)$ is $\left[0, \frac{\pi}{2}\right]$

$\therefore f(x) = \frac{x + \sin x}{2}$, $x \in [0, \pi]$

Now let $\pi \leq t \leq 2\pi$, then $f(t) + f(2\pi - t) = \pi$

i.e $f(t) + \frac{2\pi - t + \sin(2\pi - t)}{2} = \pi$

i.e $f(t) + \pi - \frac{t}{2} - \frac{\sin t}{2} = \pi$

i.e $f(t) = \frac{t + \sin t}{2}$

$\therefore f(x) = \frac{x + \sin x}{2}$ for $\pi \leq x \leq 2\pi$

Thus $f(x) = \frac{x + \sin x}{2}$ for $0 \leq x \leq 2\pi$

Also $f(x) = f(4\pi - x)$ for all $x \in [2\pi, 4\pi] \Rightarrow f(x)$ is symmetric about $x = 2\pi$

$\therefore$ from graph of $f(x)$

$\therefore \alpha = 2\pi - 0 = 2\pi$

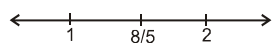
$\therefore \beta = \alpha$

Maximum value is $f(2\pi) = \pi = \frac{\beta}{2}$

7. $y = (x - 1)^3 (x - 2)^2$



$$\frac{dy}{dx} = 3(x-1)^2(x-2)^2 + 2(x-2)(x-1)^3 = (x-1)^2(x-2)[3(x-2) + 2(x-1)] = (x-1)^2(x-2)(5x-8)$$



$$(x^2 - 2x + 1)(5x^2 - 18x + 16)$$

$$\frac{d^2y}{dx^2} = (2x-2)(5x^2-18x+16) + (10x-18)(x^2-2x+1) = 0 = 20x^3 - 42x^2 + 11x - 50 = 0$$

$$= 10x^3 - 42x^2 + 57x - 25 = 0$$

$$(x-1)(10x^2 - 32x + 25) = 0$$

$$x = 1 \quad \text{or} \quad x = \frac{32 \pm \sqrt{24}}{20}$$

no. of points of inflections = 3

8. $f(x) = x^4 + ax^3 + \frac{3x^2}{2} + 1$

$$f'(x) = 4x^3 + 3ax^2 + 3x$$

$$f''(x) = 12x^2 + 6ax + 3$$

Now, $f(x)$ will be concave upward along the entire real line iff

$$f''(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$12x^2 + 6ax + 3 > 0 \quad \Rightarrow \quad D \leq 0$$

$$36a^2 - 144 \leq 0$$

$$a^2 - 4 \leq 0 \quad \Rightarrow \quad a \in [-2, 2]$$

9. $\sin x$ is concave downward in $(0, \pi)$ and $\sin x$ is concave upward in $(\pi, 2\pi)$

10. $e^x, 2^x, \tan^{-1}x$ (If $x \in \mathbb{R}^+$) is concave upward and $\ln x$ is concave downward

11. $f(x)$ concave downward ($f''(x) < 0$)

$$f(x) \text{ increasing } (\therefore f'(x) > 0)$$

$$\text{Let } g(x) = f^{-1}(x) = x$$

$$f'(g(x)) = x$$

$$f'(g(x)) \cdot g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))} > 0$$

$$g''(x) = -\frac{1}{(f'(g(x)))^2} \times f''(x) \cdot g'(x)$$

$$g''(x) > 0$$

$g(x) = f^{-1}(x)$ concave upward



EXERCISE # 3

PART - I

1. $f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$
 $f(x) = \ln(g(x))$

$$\Rightarrow g(x) = e^{f(x)} \quad \begin{array}{c} + \quad - \quad + \\ 2009 \quad 2011 \end{array}$$

$$\Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$$

only point of maxima [Applying first derivative test]

2. Clearly $f(x) = e^{x^2} + e^{-x^2}$

$$f'(x) = 2x(e^{x^2} - e^{-x^2}) \geq 0 \text{ increasing} \Rightarrow f_{\max} = f(1) = e + \frac{1}{e}$$

$$g(x) = xe^{x^2} + e^{-x^2} \Rightarrow g'(x) = e^{x^2} + 2x^2e^{x^2} - 2xe^{-x^2} > 0 \text{ increasing}$$

$$\Rightarrow g_{\max} = g(1) = e + \frac{1}{e}$$

$$h(x) = x^2e^{x^2} + e^{-x^2} \Rightarrow h'(x) = 2xe^{x^2} + 2x^3e^{x^2} - 2xe^{-x^2} = 2x(e^{x^2} + x^2e^{x^2} - e^{-x^2}) > 0$$

$$\Rightarrow h_{\max} = h(1) = e + \frac{1}{e}, \text{ so } a = b = c$$

3. (A) $\operatorname{Re} \left(\frac{2i(x+iy)}{1-(x^2-y^2+2xyi)} \right) = \operatorname{Re} \left(\frac{-2y+2ix}{1-x^2+y^2-2xyi} \right) = \operatorname{Re} \left(\frac{-2y+2ix}{2y(y-ix)} \right) = \operatorname{Re} \left(\frac{-1}{y} \right) = \frac{-1}{y}$
 $= -1 \leq y \leq 1 \Rightarrow \frac{-1}{y} \geq 1 \text{ or } \frac{-1}{y} \leq -1$

Alternate

$$\operatorname{Re} \left(\frac{2ie^{i\theta}}{1-e^{2i\theta}} \right) = \operatorname{Re} \left(\frac{2i(\cos\theta + i\sin\theta)}{1-(\cos 2\theta + i\sin 2\theta)} \right)$$

$$= \operatorname{Re} \left(\frac{2i(\cos\theta + i\sin\theta)}{2\sin^2\theta - 2i\sin\theta\cos\theta} \right) = \operatorname{Re} \left(\frac{i(\cos\theta + i\sin\theta)}{\sin\theta(\sin\theta - i\cos\theta)} \right) = \operatorname{Re} \left(\frac{(\cos\theta + i\sin\theta)}{-\sin\theta(\cos\theta + i\sin\theta)} \right) = \operatorname{Re} \left(\frac{-1}{\sin\theta} \right)$$

as $-1 \leq \sin\theta \leq 1 \Rightarrow (-\infty, 0) \cup (0, \infty)$

(B) $-1 \leq \frac{8 \cdot 3^{x-2}}{1-3^{2x-2}} \leq 1 \Rightarrow -1 \leq \frac{8t}{9-t^2} \leq 1$

$$\Rightarrow -1 \leq \frac{8t}{9-t^2} \leq 1 \Rightarrow 0 \leq \frac{9-t^2+8t}{9-t^2} \cap \frac{8t}{9-t^2} - 1 \leq 0$$

$$\Rightarrow 0 \leq \frac{t^2-8t-9}{t^2-9} \cap \frac{8t-9+t^2}{9-t^2} \leq 0 \Rightarrow 0 \leq \frac{(t-9)(t+1)}{(t-3)(t+3)} \cap \frac{(t+9)(t-1)}{(t-3)(t+3)} \geq 0$$

$$\Rightarrow t \in (-\infty, -9] \cup [-1, 1] \cup [9, \infty) \Rightarrow x \in (-\infty, 0) \cup [2, \infty)$$

Number line for t: $-\infty \dots -9 \quad -3 \quad -1 \quad 1 \quad 3 \quad 9 \dots \infty$

Number line for x: $-\infty \dots 0 \quad 2 \dots \infty$

(C) $\Rightarrow t \in (-\infty, -9] \cup [-1, 1] \cup [9, \infty) \Rightarrow x \in (-\infty, 0) \cup [2, \infty)$
 $f(\theta) = 2 \sec^2\theta \Rightarrow f(\theta) \geq 2 \Rightarrow f(\theta) \in [2, \infty)$

(D) $f(x) = x^{3/2}(3x-10) \Rightarrow f'(x) = x^{3/2} \cdot 3 + \frac{3}{2} x^{1/2}(3x-10)$
as $f'(x) \geq 0 \Rightarrow x^{1/2} \left[3x + \frac{3}{2}(3x-10) \right] \geq 0 \Rightarrow 3x + \frac{9x}{2} - 15 \geq 0$
 $\Rightarrow \frac{15x}{2} - 15 \geq 0 \Rightarrow x \geq 2 \Rightarrow x \in [2, \infty)$

4. $f(x) = x^4 - 4x^3 + 12x^2 + x - 1$

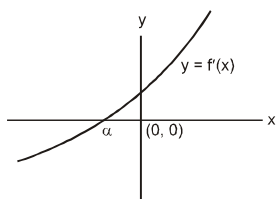




$$f'(x) = 4x^3 - 12x^2 + 24x + 1$$

$$f''(x) = 12x^2 - 24x + 24 = 12(x^2 - 2x + 2) > 0 \quad x \in \mathbb{R}$$

$\therefore f'(x)$ is S.I. function



Let α is a real root of the equation $f'(x) = 0$

$\therefore f(x)$ is MD for $x \in (-\infty, \alpha)$ and M.I. for $x \in (\alpha, \infty)$ where $\alpha < 0$

$\therefore f(0) = -1$ and $\alpha < 0 \Rightarrow f(\alpha)$ is also negative

$\therefore f(x) = 0$ has two real & distinct roots.

5. $p' = \lambda(x-1)(x-3) = \lambda(x^2 - 4x + 3)$

$$p(x) = \lambda(x^3/3 - 2x^2 + 3x) + \mu$$

$$p(1) = 6$$

$$6 = \lambda(1/3 - 2 + 3) + \mu$$

$$6 = \lambda(1/3 + 1) + \mu$$

$$18 = 4\lambda + 3\mu \quad \dots(i)$$

$$p(3) = 2$$

$$2 = \lambda(27/3 - 2 \times 9 + 9) + \mu$$

$$2 = \mu$$

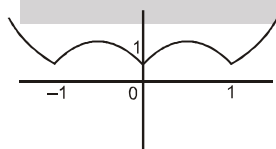
$$\mu = 2 \Rightarrow \lambda = 3$$

$$p'(x) = 3(x-1)(x-3)$$

$$p'(0) = 3(-1)(-3) = 9$$

6. $f(x) = |x| + |x^2 - 1|$

$$f(x) = \begin{cases} -x + x^2 - 1 & x < -1 \\ -x - x^2 + 1 & -1 \leq x \leq 0 \\ x - x^2 + 1 & 0 < x < 1 \\ x + x^2 - 1 & x \geq 1 \end{cases}$$



$$f(x) = \begin{cases} x^2 - x - 1 & x < -1 \\ -x^2 - x + 1 & -1 \leq x \leq 0 \\ -x^2 + x + 1 & 0 < x < 1 \\ x^2 + x - 1 & x \geq 1 \end{cases}$$



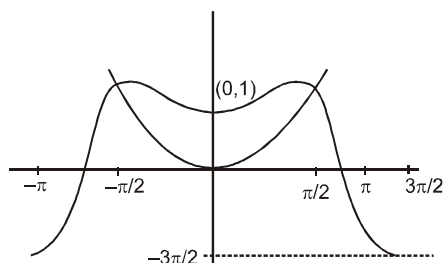
7. $x^2 = x \sin x + \cos x$

$$f(x) = x^2$$

$$g(x) = x \sin x + \cos x$$

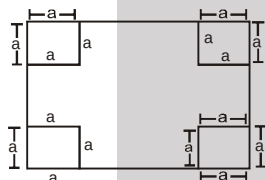
$$g'(x) = \sin x + x \cos x - \sin x$$

$$g'(x) = x \cos x$$



Only two solution.

8*



Let $\ell = 8x$, $b = 15x$

$$\therefore \text{Volume} = (8x - 2a)(15x - 2a)(a) = 4a^3 - 46a^2x + 120ax^2$$

$$\frac{dV}{da} = 6a^2 - 46ax + 60x^2$$

$$\left(\frac{dV}{da}\right)_{\text{at } x=5} = 0$$

$$\therefore x = 3 \text{ and } \frac{5}{6}$$

$$\frac{d^2V}{da^2} = 6a - 23x$$

$$\left(\frac{d^2V}{da^2}\right)_{\text{at } a=5 \text{ \& } x=3} < 0,$$

So, at $x = 3$ gives maxima $\left(\frac{d^2V}{da^2}\right)_{\text{at } a=5 \text{ \& } x=\frac{5}{6}} > 0$

So, at $x = \frac{5}{6}$ gives minima. $\frac{dV}{da} = 0$ when $a = 5$ given

($\therefore 4a^2 = 100$ given for maximum volume)

at $a = 5$

by $\frac{dV}{da} = 0$

$$\Rightarrow 6x^2 - 23x + 15 = 0$$

$$x = 3 \text{ or } \frac{5}{6}$$

So by $x = 3$ (for max volume)

$$8x = 24, 15x = 45$$

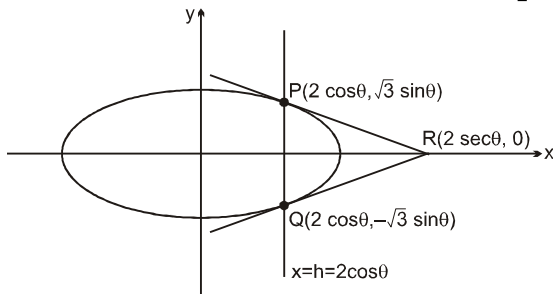




9. Point of intersection of tangents at P and Q is R(2 secθ, 0)

$$\text{Area of } \Delta PQR = \frac{1}{2} \cdot 2\sqrt{3} \sin \theta \cdot (2 \sec \theta - 2 \cos \theta)$$

$$\Rightarrow \Delta = 2\sqrt{3} \cdot \frac{\sin^3 \theta}{\cos \theta} ; \text{ where } \cos \theta \in \left[\frac{1}{4}, \frac{1}{2} \right]$$



$$\text{Now } \frac{d\Delta}{d\theta} = \frac{2\sqrt{3} [\cos \theta \cdot 3 \sin^2 \theta \cos \theta - \sin^3 \theta (-\sin \theta)]}{\cos^2 \theta} > 0$$

As θ increases, Δ increases $\Rightarrow$ when $\cos \theta$ decreases, Δ increases

$$\therefore \Delta_{\min.} \text{ occurs at } \cos \theta = 1/2, \text{ Therefore } \Delta_2 = 2\sqrt{3} \cdot \frac{(1-1/4)^{3/2}}{1/2} = 4\sqrt{3} \cdot \frac{3\sqrt{3}}{8} = \frac{36}{8}$$

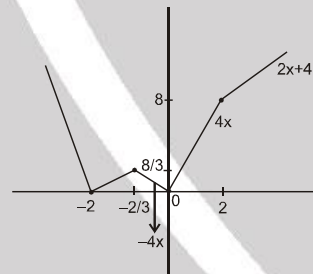
$$\Delta_{\max.} \text{ occurs at } \cos \theta = 1/4, \text{ Therefore } \Delta_1 = 2\sqrt{3} \cdot \frac{(1-1/16)^{3/2}}{1/4} = 8\sqrt{3} \cdot \frac{15\sqrt{15}}{4 \cdot 4 \cdot 4} = \frac{2\sqrt{3} \cdot 15 \cdot \sqrt{3} \sqrt{5}}{16}$$

$$\Rightarrow \Delta_1 = \frac{45}{8} \sqrt{5}$$

$$\text{Now } \frac{8}{\sqrt{5}} \Delta_1 - 8\Delta_2 = 45 - 36 = 9$$

10*. $f(x) = 2|x| + |x+2| - ||x+2| - 2|x||$

$$= \begin{cases} -2x-4 & x \leq -2 \\ 2x+4 & -2 < x \leq -2/3 \\ -4x & -2/3 < x \leq 0 \\ 4x & 0 < x \leq 2 \\ 2x+4 & x > 2 \end{cases}$$



Graph of $y = f(x)$ is minima at $x = -2, 0$; maxima at $x = -2/3$

11. $f''(x) - 2f'(x) + f(x) \geq e^x$

$$f''(x) \cdot e^{-x} - f'(x)e^{-x} - f'(x)e^{-x} + f(x)e^{-x} \geq 1$$

$$\frac{d}{dx} (f'(x)e^{-x}) - \frac{d}{dx} (f(x) \cdot e^{-x}) \geq 1$$

$$\frac{d}{dx} (f'(x) e^{-x} - f(x) e^{-x}) \geq 1 \Rightarrow \frac{d^2}{dx^2} (e^{-x}f(x)) \geq 1 \quad \forall x \in [0, 1]$$

$$\text{Let } \phi(x) = e^{-x}f(x)$$

$$\Rightarrow \phi(x) \text{ is concave upward } f(0) = f(1) = 0$$



$$\Rightarrow \phi(0) = 0 = \phi(1) \Rightarrow \phi(x) < 0 \Rightarrow f(x) < 0$$

12. $\phi'(x) < 0, x \in (0, 1/4)$ and $\phi'(x) > 0, x \in (1/4, 1) \Rightarrow e^{-x} f'(x) - e^{-x} f(x) < 0, x \in (0, 1/4)$
 $f'(x) < f(x), 0 < x < 1/4$

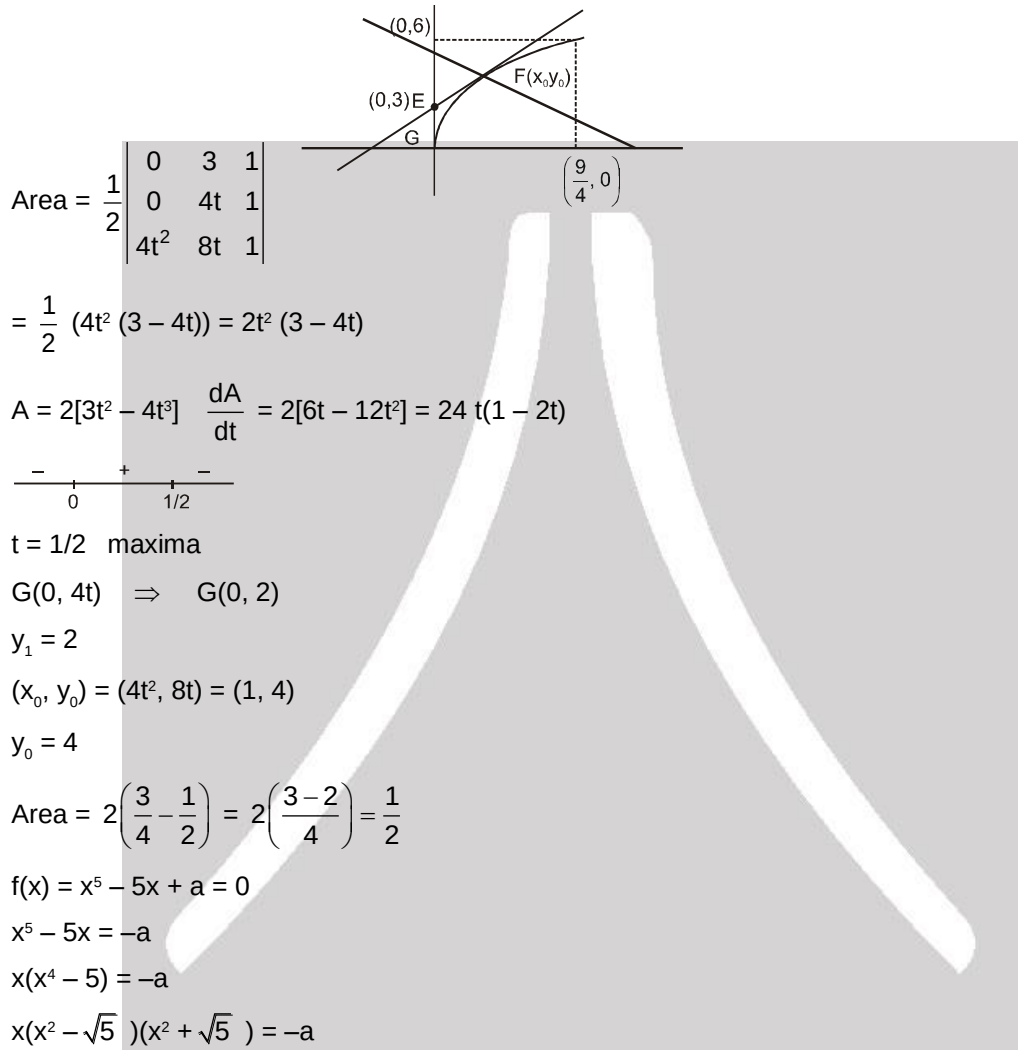
13. tangent at $Fy = x + 4t^2$

$$a : x = 0y = 4t \quad (0, 4t)$$

$$(4t^2, 8t) \text{ satisfies the line}$$

$$8t = 4mt^2 + 3$$

$$4mt^2 - 8t + 3 = 0$$



14*. $f(x) = x^5 - 5x + a = 0$

$$x^5 - 5x = -a$$

$$x(x^4 - 5) = -a$$

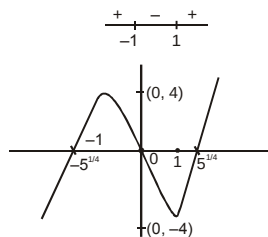
$$x(x^2 - \sqrt{5})(x^2 + \sqrt{5}) = -a$$

$$x(x - 5^{1/4})(x + 5^{1/4})(x^2 + \sqrt{5}) = -a \quad \dots(1)$$

$$f'(x) = 5x^4 - 5 = 0$$

$$(x^2 - 1)(x^2 + 1) = 0$$

$$(x - 1)(x + 1)(x^2 + 1) = 0$$







15. $(y - x^5)^2 = x(1 + x^2)^2$

$$2(y - x^5) \left(\frac{dy}{dx} - 5x^4 \right) = (1 + x^2)^2 + 2x(1 + x^2) \cdot 2x$$

at point (1, 3)

$$\therefore 2(3 - 1) \left(\frac{dy}{dx} - 5 \right) = 4 + 8$$

$$\frac{dy}{dx} - 5 = \frac{12}{4} = 3$$

$$\frac{dy}{dx} = 8$$

16. Volume of material $V = \pi r^2 h$

$$\Rightarrow V_1 = \pi(r + 2)^2 h + \pi(r + 2)^2 h - \pi r^2 h$$

$$\Rightarrow V_1 = 2\pi(r + 2)^2 + \pi h(4 + 4r)$$

$$\Rightarrow V_1 = 2\pi(r + 2)^2 + 4\pi h(r + 1)$$

$$\Rightarrow V_1 = 2\pi \left((r + 2)^2 + \frac{2(r + 1)V}{\pi r^2} \right)$$

$$\Rightarrow \frac{dV_1}{dr} = 2\pi \left(2(r + 2) + \frac{2V}{\pi} \left(\frac{-1}{r^2} - \frac{2}{r^3} \right) \right) = 0$$

$$\Rightarrow 24 + \frac{2V}{\pi} \left(\frac{-2 - 10}{10^3} \right) = 0$$

$$\Rightarrow \frac{24V}{10^3 \pi} = 24 \Rightarrow V = 10^3 \pi$$

$$\Rightarrow \frac{V}{250\pi} = 4$$

17*. Let $h(x) = f(x) - 3g(x)$

$$\left. \begin{aligned} h(-1) &= 3 \\ h(0) &= 3 \end{aligned} \right\}$$

$\Rightarrow h'(x) = 0$ has atleast one root in $(-1, 0)$ and atleast one root in $(0, 2)$

$$h(2) = 3$$

But since $h''(x) = 0$ has no root in $(-1, 0)$ & $(0, 2)$ therefore $h'(x) = 0$ has exactly 1 root in $(-1, 0)$ & exactly 1 root in $(0, 2)$

18. $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$

$$\therefore \lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} \left(\frac{0}{0} \right) \text{ Indeterminant form as } f(2) = 0, g(2) = 0$$

$\therefore$ Using L.H.

$$\lim_{x \rightarrow 2} \frac{f'(x)g(x) + g'(x)f(x)}{f''(x)g'(x) + g''(x)f'(x)} = \frac{f'(2)g(2) + g'(2)f(2)}{f''(2)g'(2) + g''(2)f'(2)} = \frac{g'(2)f(2)}{f''(2)g'(2)} = 1 \Rightarrow f''(2) = f(2)$$

and $f'(2) = 0$ & range of $f(x) \in (0, \infty)$ so $f''(2) = f(2) = +ve$

so $f(x)$ has point of minima at $x = 2$

and $f(2) = f''(2)$ so $f(x) = f''(x)$ have atleast one solution in $x \in \mathbb{R}$



(19 to 21)

$$f(x) = x + \ln x - x \ln x$$

$$f'(x) = 1 + \frac{1}{x} - \ln x - x \left(\frac{1}{x} \right) = \frac{1}{x} - \ln x$$

$$f''(x) = -\frac{1}{x^2} - \frac{1}{x} < 0 \quad \forall x \in (0, \infty)$$

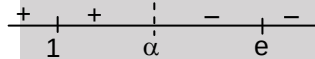
$\therefore f'(x)$ is strictly decreasing function for $x \in (0, \infty)$

$$\left. \begin{array}{l} \lim_{x \rightarrow \infty} f'(x) = -\infty \\ \lim_{x \rightarrow 0^+} f'(x) = \infty \end{array} \right\} \Rightarrow f'(x) = 0 \text{ has only one real root in } (0, \infty)$$

$$f'(1) = 1 > 0$$

$$f'(e) = \frac{1}{e} - 1 < 0 \quad \therefore f'(x) = 0 \text{ has one root in } (1, e)$$

Let $f'(\alpha) = 0$, where $\alpha \in (1, e)$



$\therefore f(x)$ is increasing in $(0, \alpha)$ and decreasing in (α, ∞)

$$f(1) = 1 \text{ and } f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$$

$$\Rightarrow f(x) = 0 \text{ has one root in } (1, e^2)$$

From column 1 : I and II are correct.

From column 2 : ii, iii, and iv are correct.

From column 3 : P, Q, S are correct

22. $f''(x) > 0$ for all $x \in \mathbb{R}$, $f(1/2) = 1/2$, $f(1) = 1$

$\Rightarrow f'(x)$ increases

$$\text{Let } g(x) = f(x) - x, \quad x \in [1/2, 1]$$

Then $g'(x) = 0$ has at least one real root in $(1/2, 1)$

$f'(x) = 1$ has at least one real root in $(1/2, 1)$

Hence $f'(x)$ increases $\Rightarrow f'(1) > 1$

23. $f'(x) - 2f(x) > 0$

$$\Rightarrow \frac{d}{dx} (f(x) \cdot e^{-2x}) > 0$$

$\Rightarrow g(x) = f(x) \cdot e^{-2x}$ is an increasing function. for $x > 0$, $g(x) > g(0)$

$$\Rightarrow f(x) \cdot e^{-2x} > 1$$

$$\Rightarrow f(x) > e^{2x}$$

$$\text{Now } f'(x) > 2f(x) > 2 \cdot e^{2x}$$

$\therefore f(x)$ is an increasing function



$$24. \quad f(x) = \begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = \cos 2x - \cos 2x (-\cos^2 x + \sin^2 x) + \sin 2x (-2\sin x \cos x)$$

$$f(x) = \cos 4x + \cos 2x$$

$$\therefore f(x) = 2\cos^2 2x + \cos 2x - 1$$

$$\text{Let } \cos 2x = t$$

$$\Rightarrow f(x) = 2t^2 + t - 1 \text{ and } t \in [-1, 1]$$

$$f(x) \text{ attains its minima at } t = -\frac{1}{4} \in [-1, 1]$$

$$f(x), t = -\frac{1}{4} \in [-1, 1]$$

$$\therefore f(x)_{\min} = \frac{2}{16} - \frac{1}{4} - 1 = \frac{-9}{8}$$

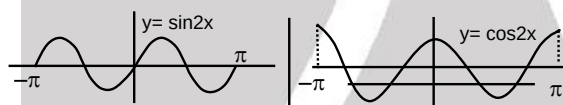
$$\therefore f(x)_{\max} = 2 + 1 - 1 = 2 \dots \dots \dots (\text{when } \cos 2x = 1)$$

$$f'(x) = -4\sin 4x - 2\sin 2x$$

$$f'(x) = 0 \Rightarrow 4\sin 4x + 2\sin 2x = 0$$

$$\Rightarrow 8\sin 2x \cos 2x + 2\sin 2x = 0 \Rightarrow 2\sin 2x (4\cos 2x + 1) = 0$$

$$\Rightarrow \sin 2x = 0 \text{ or } \cos 2x = -\frac{1}{4}$$



Hence option (B), (C)

$$25. \quad f^2(0) + (f'(0))^2 = 85 \quad f: \mathbb{R} \rightarrow [-2, 2]$$

(A) This is true of every continuous function

$$(B) \quad f'(c) = \frac{f(-4) - f(0)}{-4 - 0}$$

$$|f'(c)| = \left| \frac{f(-4) - f(0)}{4} \right|$$

$$-2 \leq f(-4) \leq 2$$

$$-2 \leq f(0) \leq 2$$

$$-4 \leq f(-4) - f(0) \leq 4$$

$$\text{This } |f'(c)| \leq 1$$

$$(C) \quad \lim_{x \rightarrow \infty} f(x) = 1$$

Note $f(x)$ should have a bound ∞ which can be concluded by considering

$$f(x) = 2 \sin \left(\frac{\sqrt{85} x}{2} \right) \quad f'(x) = \sqrt{85} \cos \left(\frac{\sqrt{85} x}{2} \right)$$

$$f^2(0) + (f'(0))^2 = 85$$

and $\lim_{x \rightarrow \infty} f(x)$ does not exist



(D) Consider $H(x) = f^2(x) + (f'(x))^2$

$$H(0) = 85$$

By (B) choice there exists some x_0 such that $(f'(x_0))^2 \leq 1$ for some x_0 in $(-4, 0)$

$$\text{hence } H(x_0) = f^2(x_0) + (f'(x_0))^2 \leq 4 + 1$$

$$H(x_0) \leq 5$$

Hence let $p \in (-4, 0)$ for which $H(p) = 5$

(note that we have considered p as largest such negative number)

similarly let q be smallest positive number $\in (0, 4)$ such that $H(q) = 5$

Hence By Rolle's theorem is (p, q)

$H'(c) = 0$ for some $c \in (-4, 4)$ and since $H(x)$ is greater than 5 as we move from $x = p$

to $x = q$ and $f^2(x) \leq 4 \Rightarrow (f'(x))^2 \geq 1$ in (p, q) Thus $H'(c) = 0 \Rightarrow f'f + f'f'' = 0$

so $f + f'' = 0$ and $f' \neq 0$

$$26. \quad f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1 & x < 0 \\ x^2 - x + 1 & 0 \leq x < 1 \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & 1 \leq x < 3 \\ (x-2)\ln(x-2) - x + \frac{10}{3} & x \geq 3 \end{cases}$$

$$f'(x) = \begin{cases} 5(x+1)^4 - 2 & x < 0 \\ 2x - 1 & 0 \leq x < 1 \\ 2x^2 - 8x + 7 & 1 \leq x < 3 \\ \ln(x-2) & x \geq 3 \end{cases}$$

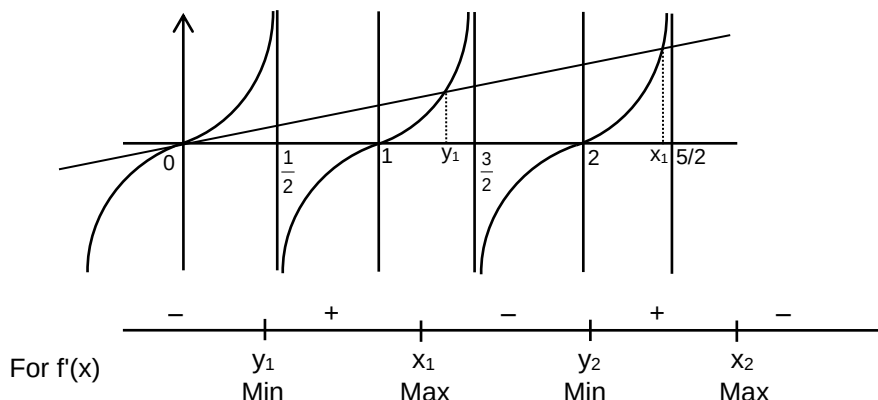
$x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1$ takes value between $-\infty$ to 1

Also $(x-2)\ln(x-2) - x + \frac{10}{3}$ takes value between $\frac{1}{3}$ to ∞

So, range of $f(x)$ is $\mathbb{R}$. So option (A) is correct $f''(1^-) = 2$ and $f''(1^+) = -4$

so $f'(x)$ is non-diff at $x = 1$ so option (B) is correct $f'(x)$ has local maxima at $x = 1$ so option (C) is correct

$$27. \quad f'(x) = \frac{2x \cos \pi x \left(\frac{\pi x}{2} - \tan \pi x \right)}{x^4}$$





PART - II

1. $\lim_{x \rightarrow -1^+} f(x) = 1$

$f(-1) = k + 2$

$\lim_{x \rightarrow (-1)^-} f(x) = k + 2$

$\therefore$ f has a local minimum at $x = -1$

$\therefore f(-1^+) \geq f(-1) \leq f(-1^-)$

$1 \geq k + 2 \leq k + 2$

$\Rightarrow k \leq -1$

possible value of k is -1

Hence correct option is (3)

2. $e^x + 2e^{-x} \geq 2\sqrt{2}$ (AM $\geq$ GM)

$\frac{1}{e^x + 2e^{-x}} \leq \frac{1}{2\sqrt{2}}$

$\frac{1}{2\sqrt{2}} \geq f(x) > 0$ so statement- 2 is correct

As $f(x)$ is continuous and $\frac{1}{3}$ belongs to range $\left(0, \frac{1}{2\sqrt{2}}\right]$ of $f(x)$,

$\Rightarrow f(c) = \frac{1}{3}$ for some C .

Hence correction option is (4).

3. $y = x + \frac{4}{x^2}$

$y' = 1 - \frac{8}{x^3} = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$

$y = 2 + \frac{4}{2^2} = 3$

(2, 3) is point of contact

Thus $y = 3$ is tangent

Hence correct option is (3)

4. $f(x) = \begin{cases} \frac{\tan x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$

In right neighbourhood of '0'

$\tan x > x$

$\frac{\tan x}{x} > 1$

In left neighbourhood of '0'

$\tan x < x$

$\frac{\tan x}{x} > 1$ as ($x < 0$)

at $x = 0$, $f(x) = 1$

$\Rightarrow x = 0$ is point of minima

so statement 1 is true.

statement 2 obvious



5. $y - x = 1$
 $y^2 = x$
 $2y \frac{dy}{dx} = 1$
 $\frac{dy}{dx} = \frac{1}{2y} = 1$
 $y = \frac{1}{2}$
 $x = \frac{1}{4}$
 tangent at $\left(\frac{1}{4}, \frac{1}{2}\right)$ $\frac{1}{2} y = \frac{1}{2} \left(x + \frac{1}{4}\right)$ $y = x + \frac{1}{4}$ $y - x = \frac{1}{4}$

distance = $\left| \frac{1 - \frac{1}{4}}{\sqrt{2}} \right| = \frac{3}{4\sqrt{2}} = \frac{3\sqrt{2}}{8}$

6. $V = \frac{4}{3} \pi r^3$ $4500 \pi = \frac{4\pi r^3}{3}$
 $\frac{dV}{dt} = 4\pi r^2 \left(\frac{dr}{dt}\right)$ $45 \times 25 \times 3 = r^3$
 $r = 15 \text{ m}$
 after 49 min = $(4500 - 49.72)\pi = 972 \pi \text{ m}^3$
 $972 \pi = \pi r^3$
 $r^3 = 3 \times 243 = 3 \times 3^5$
 $r = 9$

$72 \pi = 4\pi \times 9 \times 9 \left(\frac{dr}{dt}\right)$

$\frac{dr}{dt} = \left(\frac{2}{9}\right)$

$r^3 = 3 \times 243 = 3 \times 3^5$
 $r = 9$

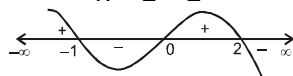
$72 \pi = 4\pi \times 9 \times 9 \left(\frac{dr}{dt}\right) \Rightarrow \frac{dr}{dt} = \left(\frac{2}{9}\right)$

7. $f'(x) = \frac{1}{x} + 2bx + a$
 at $x = -1$ $-1 - 2b + a = 0$
 $a - 2b = 1$... (i)
 at $x = 2$ $\frac{1}{2} + 4b + a = 0$

$a + 4b = -\frac{1}{2}$... (ii)

On solving (i) and (ii) $a = \frac{1}{2}$, $b = -\frac{1}{4}$

$f'(x) = \frac{1}{x} - \frac{x}{2} + \frac{1}{2} = \frac{2 - x^2 + x}{2x} = \frac{-(x+1)(x-2)}{2x}$



So maxima at $x = -1, 2$



8. $f(x) = 2x^3 + 3x + k$
 $f'(x) = 6x^2 + 3 > 0 \quad \forall x \in \mathbb{R}$
 $\Rightarrow f(x)$ is strictly increasing function
 $\Rightarrow f(x) = 0$ has only one real root, so two roots are not possible

9. Consider $f(x) - 2g(x) = h(x)$
 Then, $h(x)$ is continuous and differentiable in $[0, 1]$
 Also $h(0) = 2$ & $h(1) = 2$
 Hence $h(x)$ satisfies conditions of Rolle's Theorem in $(0, 1)$
 Thus, There exist a 'c' such that $h'(c) = 0$ where $c \in (0, 1)$
 $\Rightarrow f'(c) = 2g'(c)$

10. $f(x) = \alpha \ln|x| + \beta x^2 + x$
 (1). $f'(x) = \frac{\alpha}{x} + 2\beta x + 1 = \frac{2\beta x^2 + x + \alpha}{x}$
 Since $x = -1, 2$ are extreme points $\Rightarrow f'(x) = 0$ at these points.
 Hence $2\beta - 1 + \alpha = 0$
 $8\beta + 2 + \alpha = 0$
 $-6\beta - 3 = 0 \Rightarrow \beta = -\frac{1}{2} \text{ \& } \alpha = 2.$

11. $4x + 2\pi r = 2 \quad \dots(i)$
 $x^2 + \pi r^2 = \text{minimum} \Rightarrow \text{So } f(r) = \left(\frac{1-\pi r}{2}\right)^2 + \pi r^2$
 $\frac{df}{dr} = \pi^2 \frac{r}{2} - \frac{\pi}{2} + 2\pi r = 0 \Rightarrow r = \frac{1}{\pi + 4}$
 using equation (i) $x = \frac{(1-\pi r)}{2} \Rightarrow x = 2r$

12. at $x = \frac{\pi}{6} \Rightarrow y = \frac{\pi}{3}$
 $f(x) = \tan^{-1} \left(\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right) \quad \because x \in \left(0, \frac{\pi}{2}\right)$
 $= \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right)$
 $f(x) = \frac{\pi}{4} + \frac{x}{2} \quad f'(x) = \frac{1}{2}$
 slope of normal = -2
 equation of normal $y - \frac{\pi}{3} = -2 \left(x - \frac{\pi}{6} \right)$
 $y = -2x + \frac{2\pi}{3}$



13. $2r + \ell = 20 \Rightarrow 2r + r\theta = 20 \Rightarrow \theta = \frac{20-2r}{r} \Rightarrow A = \frac{\pi r^2 \theta}{360} = \frac{r^2}{2} \cdot \frac{20-2r}{r} = r(10-r)$

$$A = 10r - r^2 \Rightarrow \frac{dA}{dr} = 10 - 2r = 0 \Rightarrow r = 5$$

$$\therefore \theta = \frac{10}{5} = 2$$

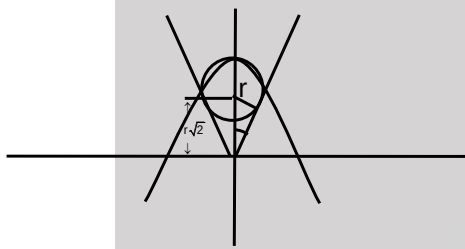
$$\therefore \text{Maximum area} = \frac{1}{2} \times 25 \times 2 = 25 \text{ sq. m.}$$

14. $y(x-2)(x-3) = x+6$

Intersection with y-axis; Put $x = 0 \Rightarrow y = 1 \Rightarrow$ Point of Intersection is $(0, 1)$

$$\text{Now, } y = \frac{x+6}{x^2-5x+6} \quad y' = \frac{(x^2-5x+6) - (x+6)(2x-5)}{(x^2-5x+6)^2} \quad y' = \frac{6-(-30)}{36} = 1 \text{ at } (0, 1)$$

$\therefore$ Equation of normal is given by $(y-1) = -1(x-0) \Rightarrow x+y-1=0$

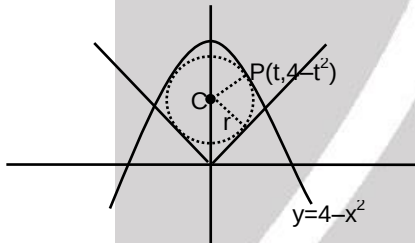


15. ✖

let radius of circle be r , its center lies on y-axis as y-axis bisects the 2 rays of $y = |x|$

$$\text{Now } 4 - r\sqrt{2} = r \Rightarrow r = \frac{4}{\sqrt{2}+1} = 4(\sqrt{2}-1)$$

NOTE : The correct solution should be



due to symmetry center of the circle must be on y-axis let center be $(0, k)$

Length of perpendicular from $(0, k)$ to $y = x$, i.e. $r = \left| \frac{k}{\sqrt{2}} \right|$

$$\therefore \text{Equation of circle : } x^2 + (y-k)^2 = \frac{k^2}{2}$$

$$\text{solving circle and parabola, } 4 - y + y^2 - 2ky + \frac{k^2}{2} = 0$$

$$y^2 - (2k+1)y + \left(\frac{k^2}{2} + 4 \right) = 0$$

Because circle touches the parabola

$\therefore D = 0$

$$(2k+1)^2 = 4 \left(\frac{k^2}{2} + 4 \right) \Rightarrow 4k^2 + 4k + 1 = 2k^2 + 16$$

$$\text{On solving we get } k = \frac{-4 + \sqrt{136}}{4}$$

Therefore radius $= k/\sqrt{2} \approx 1.3546$

However among the given choices the following method will yield one of the choice.



16. $y^2 = 6x$ and $9x^2 + by^2 = 16$

$$2y \frac{dy}{dx} = 6 \Rightarrow \frac{dy}{dx} = \frac{3}{y}$$

$$18x + 2by \frac{dy}{dx} = 0$$

$$9x + by \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-9x}{by}$$

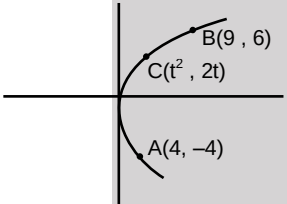
$$\frac{3}{y} \times \frac{-9x}{by} = -1$$

(b) $6x = 27x$

$$b = \frac{27}{6} \Rightarrow b = \frac{9}{2}$$

17. $f(x) = x^2 + \frac{1}{x^2}$, $g(x) = x = \frac{1}{x} \Rightarrow h(x) = \frac{f(x)}{g(x)} = \frac{x^2 + \frac{1}{x^2}}{x - \frac{1}{x}} = \frac{\left(x - \frac{1}{x}\right)^2 + 2}{\left(x - \frac{1}{x}\right)} \Rightarrow x - \frac{1}{x} = t$

$$h(t) = \frac{t^2 + 2}{t} = t + \frac{2}{t} \quad |t| \geq 2 \Rightarrow \text{AM} \geq \text{GM} \quad \frac{t + \frac{2}{t}}{2} \geq \sqrt{t \cdot \frac{2}{t}} \quad t + \frac{2}{t} \geq 2\sqrt{2}$$

18. 

$$\Delta ACB = \frac{1}{2} \begin{vmatrix} 4 & -4 & 1 \\ 9 & 6 & 1 \\ t^2 & 2t & 1 \end{vmatrix}$$

$$\Delta = 30 + 5t - 5t^2$$

$$\frac{d\Delta}{dt} = 0 \Rightarrow 5 - 10t = 0 \Rightarrow t = \frac{1}{2}$$

$$\frac{d^2\Delta}{dt^2} = -10 < 0$$

$$\therefore C\left(\frac{1}{4}, 1\right) \text{ so } \Delta = 30 + \frac{5}{2} - \frac{5}{4} = 31\frac{1}{4}$$

19. $y = 7 + x^{3/2}$

Let the point on curve be $P(x_1, 7 + x_1^{3/2})$ and given point be $A\left(\frac{1}{2}, 7\right)$

For nearest point normal at P passes through A

So slope of line AP = Slope of normal at P

$$\Rightarrow \frac{x_1^{3/2}}{x_1 - \frac{1}{2}} = -\frac{dx}{dy}\bigg|_{(x_1, y_1)} = -\frac{2}{3\sqrt{x_1}} \Rightarrow 3x_1^2 = 1 - 2x_1 \Rightarrow 3x_1^2 + 2x_1 - 1 = 0 \Rightarrow (x_1 + 1)(3x_1 - 1) = 0$$

$$\Rightarrow x_1 = \frac{1}{3} \quad (x_1 = -1 \text{ is not possible as } x_1 > 0) \Rightarrow \text{Hence point P is } \left(\frac{1}{3}, 7 + \frac{1}{3\sqrt{3}}\right)$$

$$\text{So AP} = \sqrt{\frac{1}{36} + \frac{1}{27}} = \frac{1}{6}\sqrt{\frac{7}{3}}$$



$$20. \quad f'(x) = \frac{\sqrt{a^2 + x^2} - \frac{x^2}{\sqrt{a^2 + x^2}}}{(a^2 + x^2)} - \frac{-\sqrt{b^2 + (d-x)^2} + \frac{(d-x)^2}{\sqrt{b^2 + (d-x)^2}}}{b^2 + (d-x)^2} = \frac{a^2}{(a^2 + x^2)^{3/2}} + \frac{b^2}{(b^2 + (d-x)^2)^{3/2}}$$

Hence $f(x)$ is increasing.

$$21. \quad f''(x) > 0, y = f(x); x \in (0, 2)$$

$$\phi(x) = f(x) + f(2-x)$$

$$\phi'(x) = f'(x) - f'(2-x)$$

for $\phi(x)$ to be increasing

$$\phi'(x) > 0$$

$$\Rightarrow f'(x) > f'(2-x)$$

$$\Rightarrow x > 2-x \quad (f'(x) \text{ is increasing in } (0, 2))$$

$$\Rightarrow x > 1$$

$$\Rightarrow x \in (1, 2)$$

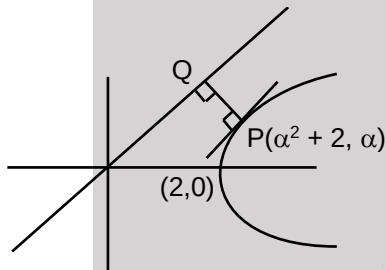
For $\phi(x)$ to be decreasing

$$\phi'(x) < 0$$

$$\Rightarrow f'(x) < f'(2-x)$$

$$\therefore x \in (0, 1)$$

22.



Shortest distance between $y^2 = x - 2$ and $y = x$

$\frac{dy}{dx}$ at point P will be 1. Differentiating the curve

$$2yy' = 1 \quad \Rightarrow \quad y' = \frac{1}{2y} = \frac{1}{2\alpha} = 1 \quad \therefore \quad P\left(\frac{9}{4}, \frac{1}{2}\right)$$

$$\therefore \text{minimum distance} = PQ = \left| \frac{\frac{9}{4} - \frac{1}{2}}{\sqrt{2}} \right| = \frac{7}{4\sqrt{2}}$$

$$23. \quad f(x) = x \sqrt{kx - x^2}$$

$$f'(x) = \sqrt{kx - x^2} + \frac{(k-2x)x}{2\sqrt{kx - x^2}} = \frac{2(kx - x^2) + kx - 2x^2}{2\sqrt{kx - x^2}} = \frac{3kx - 4x^2}{2\sqrt{kx - x^2}} = \frac{x(3k - 4x)}{2\sqrt{kx - x^2}}$$

for increasing function

$$\text{for } f'(x) \geq 0 \quad \forall x \in [0, 3]$$

$$\Rightarrow kx - x^2 \geq 0, \quad \forall x \in [0, 3] \quad \text{and} \quad x(3k - 4x) \geq 0, \quad \forall x \in [0, 3]$$

$$\Rightarrow x(x - k) \leq 0, \quad \forall x \in [0, 3] \quad \text{and} \quad x(4x - 3k) \leq 0, \quad \forall x \in [0, 3]$$

$$k \geq 3 \quad \text{and} \quad k \geq 4 \quad \Rightarrow k \geq 4 \Rightarrow m = 4$$

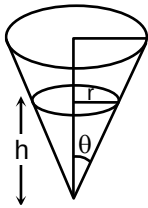
$$\text{maximum}(f(x)) \text{ when } k = 4 \text{ is } 3\sqrt{4 \times 3 - 3^2} = 3\sqrt{3} = M$$

$$(m, M) = (4, 3\sqrt{3})$$





24. $\frac{dv}{dt} = 5\text{m}^3 / \text{min} \Rightarrow v = \frac{1}{3}\pi r^2 h$



$$\tan \theta = \frac{r}{h} = \frac{1}{2} \Rightarrow 2r = h \Rightarrow v = \frac{1}{3}\pi \frac{h^3}{4} = \frac{\pi h^3}{12} \Rightarrow \frac{dv}{dt} = \frac{\pi}{4} h^2 \frac{dh}{dt} \quad 5 = \frac{\pi}{4} 10^2 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{20}{100\pi} = \frac{1}{5\pi} \text{ m/min}$$

25. $f(x) = ax^5 + bx^4 + cx^3 \quad \lim_{x \rightarrow 0} \left(2 + \frac{ax^5 + bx^4 + cx^3}{x^3} \right) = 4 \Rightarrow 2 + c = 4 \Rightarrow c = 2$

$$f'(x) = 5ax^4 + 4bx^3 + 6x^2 = x^2 (5ax^2 + 4bx + 6)$$

$$f'(1) = 0 \Rightarrow 5a + 4b + 6 = 0$$

$$f'(-1) = 0 \Rightarrow 5a - 4b + 6 = 0$$

$$b = 0$$

$$a = -\frac{6}{5} \quad f(x) = \frac{-6}{5}x^5 + 2x^3 \Rightarrow f'(x) = -6x^4 + 6x^2 = 6x^2(-x^2 + 1) = -6x^2(x+1)(x-1)$$

$$\frac{-1}{1-} \quad + \quad \frac{1-}{1}$$

Minimal at $x = -1 \Rightarrow$ Maxima at $x = 1$

26. $f'(x) = x(\pi - \cos^{-1}(\sin|x|)) = x\left(\pi - \left(\frac{\pi}{2} - \sin^{-1}(\sin|x|)\right)\right) = x\left(\frac{\pi}{2} + |x|\right)$

$$f(x) = \begin{cases} x\left(\frac{\pi}{2} + x\right) & x \geq 0 \\ x\left(\frac{\pi}{2} - x\right) & x < 0 \end{cases} \quad f'(x) = \begin{cases} \frac{\pi}{2} + 2x & x \geq 0 \\ \frac{\pi}{2} - 2x & x < 0 \end{cases}$$

$f'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ and decreasing in $\left(\frac{\pi}{2}, 0\right)$

27. $f(3) = f(4) \Rightarrow \alpha = 12 \quad f'(x) = \frac{x^2 - 12}{x(x^2 + 12)}$

$$\therefore f'(c) = 0$$

$$\therefore c = \sqrt{12}$$

$$\therefore f''(c) = \frac{1}{12}$$

28. Lets use LMVT for $x \in [a, c] \quad \frac{f(c) - f(a)}{c - a} = f'(\alpha), \alpha \in (a, c)$

also use LMVT for $x \in [c, b] \quad \frac{f(b) - f(c)}{b - c} = f'(\beta), \beta \in (c, b)$

$$\therefore f''(x) < 0 \Rightarrow f'(x) \text{ is decreasing } f'(\alpha) > f'(\beta) \quad \frac{f(c) - f(a)}{c - a} > \frac{f(b) - f(c)}{b - c} \quad \frac{f(c) - f(a)}{f(b) - f(c)} > \frac{c - a}{b - c}$$

($\therefore f(x)$ is increasing)



HIGH LEVEL PROBLEMS (HLP)

1. Given $x = f'(t) \sin t + f''(t) \cos t$

$$y = f'(t) \cos t - f''(t) \sin t$$

$$\text{from given equation } \frac{dx}{dt} = \{f'(t) + f'''(t)\} \cos t \text{ \& } \frac{dy}{dt} = -\{f'(t) + f'''(t)\} \sin t$$

$$\text{Velocity} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{\{f'(t) + f'''(t)\}^2 \cos^2 t + \{f'(t) + f'''(t)\}^2 \sin^2 t} = f'(t) + f'''(t)$$

2. $f'(x) = \sqrt{4ax - x^2} + \frac{x(4a - 2x)}{2\sqrt{4ax - x^2}} = \frac{6ax - 2x^2}{\sqrt{4ax - x^2}} < 0, \forall x \in (4a, 3a)$

so $f(x)$ is decreasing in $[4a, 3a]$

3. Here f is a differentiable function then f is continuous function. So by L.M.V. theorem for any $a \in (0, 4)$

$$f'(a) = \frac{f(4) - f(0)}{4 - 0} \dots\dots (1)$$

Again from mean value for any $b \in (0, 4)$

$$f(b) = \frac{f(4) + f(0)}{2} \dots\dots(2)$$

$$\text{Now multiplying (1) and (2), we get } \frac{f^2(4) - f^2(0)}{8} = f'(a) \cdot f(b) \Rightarrow f^2(4) - f^2(0) = 8f'(a) \cdot f(b)$$

4. $f'(x) = 0 \Rightarrow x = \frac{1}{a}, \frac{-2}{3a}$

since, we have a cubic polynomial with coefficient of $x^3 +ve$, minima will occur after maxima.

Case - 1 : If $a > 0$

$$\text{then } \frac{1}{a} = \frac{1}{3} \Rightarrow a = 3 \text{ also } f\left(\frac{1}{3}\right) > 0 \Rightarrow b < -\frac{1}{2}$$

Case - 2 : If $a < 0$

$$\text{then } -\frac{2}{3a} = \frac{1}{3} \Rightarrow a = -2$$

$$\text{also } f\left(\frac{1}{3}\right) > 0 \Rightarrow \frac{(-2)^2}{3^2} - \frac{(-2)}{2} \cdot \frac{1}{3^2} - 2\left(\frac{1}{3}\right) - b > 0 \Rightarrow \frac{4}{27} + \frac{1}{9} - \frac{2}{3} - b > 0 \Rightarrow b < -\frac{11}{27}$$

5. $f(k) = 3$

$$f(k+h) = a^2 - 2 + \frac{\sin h}{h} \Rightarrow \lim_{h \rightarrow 0} f(k+h) = a^2 - 1$$

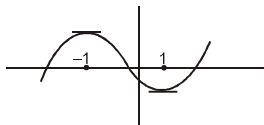
$$\lim_{h \rightarrow 0} f(k-h) = \lim_{h \rightarrow 0} (3 + |k-h-k|) = \lim_{h \rightarrow 0} (3 + |-h|) = 3 \Rightarrow a^2 - 1 > 3$$

$$a^2 > 4 \Rightarrow |a| > 2$$

6. $f(x) = x^3 - 3x + k, k = [a]$

$$f'(x) = 3(x-1)(x+1)$$

-1 is maxima is 1 is minima



Figure

$$\text{for three roots } f(-1)f(1) < 0 \Rightarrow (k+2)(k-2) < 0$$

$$k \in (-2, 2) \Rightarrow -2 < [a] < 2 \Rightarrow -1 \leq a < 2$$



7. $f(x) = \sin \frac{\{x\}}{a} + \cos \frac{\{x\}}{a} \quad a > 0$

it attains max. if $\frac{\{x\}}{a} = \frac{\pi}{4}$

$$\frac{\{x\}}{a} \in \left[0, \frac{1}{a}\right) \therefore \frac{1}{a} > \frac{\pi}{4}, \text{ for } f \text{ to have is maxima} \Rightarrow 0 < a < \frac{4}{\pi}$$

8. Let $f(x) = x^4 + 4x^3 - 8x^2 + k$

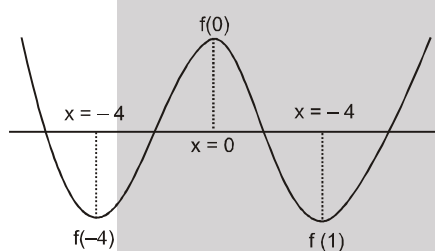
$$f'(x) = 4x^3 + 12x^2 - 16x = 4x(x^2 + 3x - 4) = 4x(x+4)(x-1) \Rightarrow f'(x) = 0 \Rightarrow x = -4, 0, 1$$

$$f''(x) = 12x^2 + 24x - 16 = 4(3x^2 + 6x - 4)$$

$$f''(-4) = 20 > 0$$

$$f''(0) = -16 < 0$$

$f''(1) = 20 > 0 \Rightarrow x = -4$ and $x = 1$ are points of local minima whereas



Figure

$x = 0$ is point of local maxima

for $f(x) = 0$ to have 4 real roots

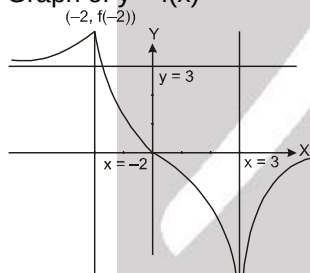
$$f(-4) < 0 \Rightarrow k < 128$$

$$f(0) > 0 \Rightarrow k > 0$$

$$f(1) < 0 \Rightarrow k < 3 \Rightarrow k \in (0, 3)$$

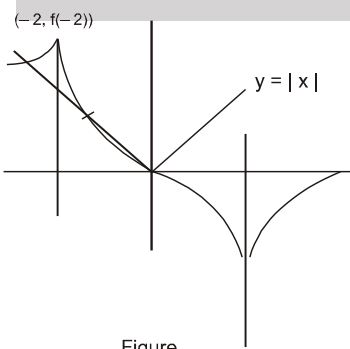
(09 to 11)

Graph of $y = f(x)$



Figure

9.

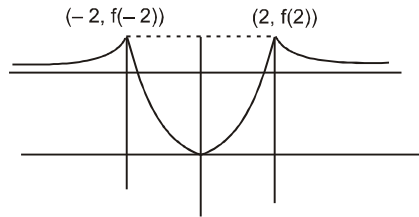


Figure

Three points of intersection. Three solutions

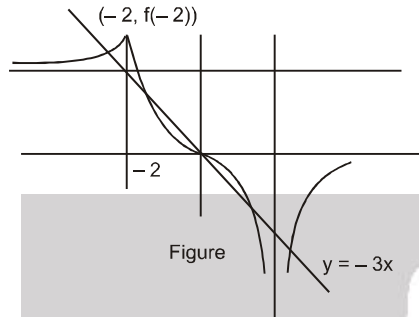


10.



Figure

11.



Figure

12.

$$(i) \quad \frac{f(-1) - f(0)}{-1 - 0} = f'(\alpha), \text{ where } -1 < \alpha < 0$$

$$\Rightarrow |f'(\alpha)| = |f(0) - f(-1)| \leq |f(0)| + |f(-1)| \Rightarrow |f'(\alpha)| \leq 1 + 1 = 2$$

similarly for $0 < \beta < 1$

$$(ii) \quad \left. \begin{array}{l} |f(x)| \leq 1 \Rightarrow (f(x))^2 \leq 1 \\ |f'(x)| \leq 2 \Rightarrow (f'(x))^2 \leq 4 \end{array} \right\} \Rightarrow F(x) \leq 5$$

(iii) Obvious from (i) and (ii) that there exists at least one max.

(iv) Also from (i) and (ii) option iv is quite obvious.

13.

$$\text{As } (a, b) \text{ lies on } y = x^2 + 1 \Rightarrow b = a^2 + 1$$

$$\left. \frac{dy}{dx} \right|_{(a, b)} = 2a$$

$$\text{Tangent } y - a^2 - 1 = 2a(x - a)$$

$$x = 0 \Rightarrow y = 1 - a^2$$

$$x = 1 \Rightarrow y = -a^2 + 2a + 1$$

$$\text{Area} = \frac{1}{2} (1) (1 - a^2 - a^2 + 2a + 1) = -a^2 + a + 1$$

$$\text{It is greatest when } a = \frac{1}{2} \Rightarrow b = 1 + \frac{1}{4} = \frac{5}{4}$$

14.

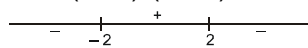
$$(2, -1) \Rightarrow -1 = \frac{2a+b}{(+1)(-2)} \Rightarrow 2a+b=2$$

$$y' = \frac{a(x-1)(x-4) - (ax+b)(2x-5)}{(x-1)^2(x-4)^2}$$

$$y' = 0 \text{ at } x = 2 \Rightarrow b = 0 \Rightarrow a = 1$$

$$y = \frac{x}{(x-1)(x-4)}$$

$$y' = \frac{(2+x)(2-x)}{(x-1)^2(x-4)^2}$$

signs of y' At $x = 2$, y' changes sign from positive to negative $\Rightarrow x = 2$ is point of maxima.



15. Since $\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \{s(s-a)(s-b)(s-c)\}^{\frac{1}{2}}$

Taking logarithm of both sides, we get $\ln \Delta = \frac{1}{2} \{ \ln s + \ln(s-a) + \ln(s-b) + \ln(s-c) \}$

$$\therefore \frac{1}{\Delta} \frac{d\Delta}{dc} = \frac{1}{2} \left\{ \frac{1}{s} \cdot \frac{ds}{dc} + \frac{1}{(s-a)} \cdot \frac{d(s-a)}{dc} + \frac{1}{(s-b)} \cdot \frac{d(s-b)}{dc} + \frac{1}{(s-c)} \cdot \frac{d(s-c)}{dc} \right\} \quad \dots\dots\dots(1)$$

But $s = \frac{1}{2}(a+b+c)$ $\frac{ds}{dc} = \frac{1}{2} \Rightarrow \frac{d(s-a)}{dc} = \frac{ds}{dc} - \frac{da}{dc} = \frac{1}{2} - 0 = \frac{1}{2}$,

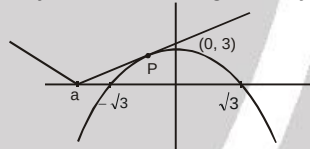
and $\frac{d(s-b)}{dc} = \frac{ds}{dc} - \frac{db}{dc} = \frac{1}{2} - 0 = \frac{1}{2}$ and $\frac{d(s-c)}{dc} = \frac{ds}{dc} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$

Now from (1), $\frac{1}{\Delta} \cdot \frac{d\Delta}{dc} = \frac{1}{2} \left\{ \frac{1}{s} \cdot \frac{1}{2} + \frac{1}{(s-a)} \cdot \frac{1}{2} + \frac{1}{(s-b)} \cdot \frac{1}{2} - \frac{1}{(s-c)} \cdot \frac{1}{2} \right\} = \frac{1}{4} \left\{ \frac{1}{s} + \frac{1}{(s-a)} + \frac{1}{(s-b)} - \frac{1}{(s-c)} \right\}$

Hence $d\Delta = \frac{\Delta}{4} \left\{ \frac{1}{s} + \frac{1}{s-a} + \frac{1}{s-b} - \frac{1}{s-c} \right\} dc$

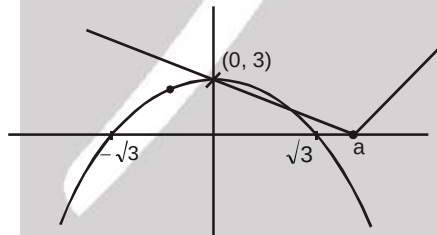
16. $3 - x^2 > |x - a|$

Case (i) $a < 0$ and $y = x - a$ is tangent of $y = 3 - x^2$ (see figure)



$-2x = 1 \Rightarrow x = -\frac{1}{2}$ P $\left(-\frac{1}{2}, \frac{11}{4}\right)$

Since $y = x - a$ passes through $\left(-\frac{1}{2}, \frac{11}{4}\right) \Rightarrow a = x - y = -\left(\frac{11}{4} + \frac{1}{2}\right) = -\frac{13}{4}$ (minimum value of a)



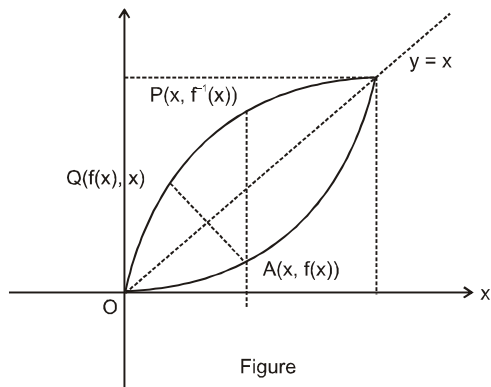
Case (ii) $a > 0$ and $y = -x + a$ passes through $(0, 3)$, then $a = 3$ (maximum value of a) (see figure)

$\Rightarrow a \in \left(-\frac{13}{4}, 3\right)$

17. Let $f(x) = x^m + a_1 x^{m-1} + a_2 x^{m-2} + \dots + a_0$ if possible, let $f(x) = 0$ has ' m ' real roots, then by Roll's theorem, $f'(x) = 0$ must have " $(m-1)$ " real roots, $f''(x) = 0$ must have " $(m-2)$ " real roots and so on, $f^{m-2}(x) = 0$ must have 2 real roots, $\frac{m!}{2} x^2 + a_1 (m-1)! x + a_2 (m-2)! = 0$ must have 2 real roots
- or $\frac{m(m-1)}{2} x^2 + a_1 (m-1) + a_2 = 0$ must have 2 real roots
- $D = a_1^2 (m-1)^2 - 2m(m-1)a_2 = (m-1)[(m-1)a_1^2 - 2a_2]$
- which is $-ve$, so our assumption is wrong. Hence proved.



18.



$$\text{Slope of OQ} > \text{slope of OP} \quad \frac{x}{f(x)} > \frac{f^{-1}(x)}{x} \Rightarrow f(x) \cdot f^{-1}(x) < x^2$$

19.

$$g'(x) = 2f'\left(\frac{x^2}{2}\right) \cdot \frac{2x}{2} + f'\left(\frac{27}{2} - x^2\right) (-2x) = x \left[f'\left(\frac{x^2}{2}\right) - f'\left(\frac{27}{2} - x^2\right) \right]$$

$$g'(x) = 0$$

$$\Rightarrow x = 0 \text{ or } \frac{x^2}{2} = \frac{27}{2} - x^2$$

$$\Rightarrow x = -3, 0, 3$$

$$\text{for } g'(x) \quad \begin{array}{cccc} - & + & - & + \\ -3 & 0 & 3 & \end{array}$$

so $g(x)$ is increasing in $x \in (-\infty, -3]$ and in $[0, 3]$ and $g(x)$ is decreasing in $[-3, 0]$ and in $[3, \infty)$

20.

$$\text{Let } f(x) = \ln x \quad \Rightarrow \quad f''(x) = -\frac{1}{x^2}$$

$$\text{So } \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n} \leq f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right)$$

for $x_1, x_2, \dots, x_n \in \mathbb{R}^+$

$$\Rightarrow \frac{\ln(x_1) + \ln(x_2) + \dots + \ln(x_n)}{n} \leq \ln\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right)$$

$$\Rightarrow (x_1 x_2 \dots x_n)^{\frac{1}{n}} \leq \frac{x_1 + x_2 + \dots + x_n}{n} \Rightarrow \text{G.M.} \leq \text{A.M.}$$

$$\text{Again } \frac{f\left(\frac{1}{x_1}\right) + f\left(\frac{1}{x_2}\right) + \dots + f\left(\frac{1}{x_n}\right)}{n} \leq f\left(\frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n}\right)$$

$$\Rightarrow \left(\frac{1}{x_1 x_2 \dots x_n}\right)^{\frac{1}{n}} \leq \frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n} \Rightarrow \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} (x_1 x_2 \dots x_n)^{\frac{1}{n}} \Rightarrow \text{G.M.} \leq \text{A.M.}$$



21. (i) $1 + x^2 > (x \sin x + \cos x)$

Let $f(x) = 1 + x^2 - x \sin x - \cos x$, $x \in [0, \infty)$

$f'(x) = 2x - \sin x - x \cos x + \sin x = x(2 - \cos x)$

$\Rightarrow f'(x) > 0$ for $x \in (0, \infty)$

$\Rightarrow f(x)$ is an increasing function

$\therefore x > 0$

$\Rightarrow f(x) > f(0)$

$\Rightarrow 1 + x^2 > x \sin x + \cos x$

(ii) $f(x) = \sin x - \sin 2x - 2x$

$f'(x) = \cos x - 2 \cos 2x - 2 = \cos x - 2(2 \cos^2 x - 1) - 2$

$= \cos x - 4 \cos^2 x = \cos x (1 - 4 \cos x)$, $x \in \left[0, \frac{\pi}{3}\right]$

$\Rightarrow \cos x \geq \frac{1}{2} \Rightarrow \cos x (1 - 4 \cos x) < 0$

$\therefore f'(x) < 0 \forall x \in \left[0, \frac{\pi}{3}\right]$

$f(x) \leq f(0) \Rightarrow \sin x - \sin 2x - 2x \leq 0$
 $\Rightarrow \sin x - \sin 2x \leq 2x$

(iii) $f(x) = \frac{x^2}{2} + 2x + 3 - 3e^x + xe^x$

$f'(x) = x + 2 - 3e^x + e^x + xe^x = x + 2 - 2e^x + xe^x$

$f''(x) = 1 - 2e^x + e^x + xe^x = 1 - e^x + xe^x$

$f'''(x) = -e^x + e^x + xe^x = xe^x$

$f'''(x) \geq 0 \forall x \geq 0$

$\Rightarrow f''(x) \geq f''(0) \Rightarrow f''(x) \geq 0$

$\Rightarrow f'(x) \geq f'(0) \Rightarrow f'(x) \geq 0$

$\Rightarrow f(x) \geq f(0) \Rightarrow f(x) > 0$

$\Rightarrow \frac{x^2}{2} + 2x + 3 \geq 3e^x - xe^x$

(iv) $f(x) = x \sin x - \frac{\sin^2 x}{2}$ $f'(x) = x \cos x + \sin x - \sin x \cos x = x \cos x + \sin x (1 - \cos x)$

$f'(x) > 0$ for $x \in \left(0, \frac{\pi}{2}\right) \Rightarrow f(x) > f(0)$ or $x \sin x - \frac{\sin^2 x}{2} > 0$

and $f(x) < f\left(\frac{\pi}{2}\right)$, $x \sin x - \frac{\sin^2 x}{2} < \frac{\pi}{2} - \frac{1}{2}$

$\Rightarrow x \sin x - \frac{\sin^2 x}{2} < \frac{1}{2} (\pi - 1)$



$$22. \quad f(x) = \left(1 - \frac{\sqrt{21-4b-b^2}}{b+1}\right) x^3 + 5x + \sqrt{6}$$

$$f'(x) = 3 \left(1 - \frac{\sqrt{21-4b-b^2}}{b+1}\right) x^2 + 5$$

$$f(x) \text{ is increasing} \Rightarrow f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow 1 - \frac{\sqrt{21-4b-b^2}}{b+1} \geq 0 \Rightarrow \frac{\sqrt{(b+7)(3-b)}}{b+1} \leq 1$$

$$\text{Case-I} \quad \text{If } b+1 > 0, \text{ then } \frac{(b+7)(3-b)}{(b+1)^2} \leq 1 \text{ and } -7 \leq b \leq 3$$

$$\Rightarrow b \leq -5 \text{ and } b \geq 2 \Rightarrow b \in [2, 3]$$

$$\text{Case-II} \quad \text{If } b+1 < 0 \Rightarrow b \in [-7, -1]$$

$$23. \quad y = x \ln x - \frac{x^2}{2} + \frac{1}{2}$$

$$y' = 1 + \ln x - x$$

$$y'' = \frac{1}{x} - 1 = \frac{1-x}{x}$$

$$y'' > 0 \quad \forall x \in (0, 1)$$

$$\Rightarrow y'(x) < y'(1)$$

$$\Rightarrow y'(x) < 0$$

$$\therefore y(x) > y(1)$$

$$\Rightarrow x \log x - \frac{x^2}{2} + \frac{1}{2} > 0$$

$$\Rightarrow x \log x > \frac{x^2}{2} - \frac{1}{2}$$

$$24. \quad f'(x) = 0 \Rightarrow x = \pm \sqrt{\frac{a}{3b}}$$

$$f\left(-\sqrt{\frac{a}{3b}}\right) = \frac{-2a}{3} \sqrt{\frac{a}{3b}}$$

$$f\left(\sqrt{\frac{a}{3b}}\right) = \frac{2a}{3} \sqrt{\frac{a}{3b}}$$

$$f(-1) = b - a$$

$$f(1) = a - b$$

$$\text{Given that } \left| \frac{2a}{3} \sqrt{\frac{a}{3b}} \right| = \left| -\frac{2a}{3} \sqrt{\frac{a}{3b}} \right| = |b - a| = |a - b| = 1$$

$$\Rightarrow \frac{4a^3}{27b} = 1 \Rightarrow b = \frac{4a^3}{27} \Rightarrow a - b = 1 \Rightarrow a - \frac{4a^3}{27} = 1 \Rightarrow 4a^3 - 27a + 27 = 0$$

$$a = -3, \frac{3}{2} \Rightarrow a = -3 - 1 : b = \frac{3}{2} - 1 = -\frac{1}{2} = -4$$

$$\text{Also, } b - a = 1 \Rightarrow \frac{4a^3}{27} - a \Rightarrow 4a^3 - 27a - 27 = 0$$

$$\Rightarrow (a-3)(2a+3)^2 = 0$$

$$\Rightarrow a = 3 \Rightarrow b = 4$$

Rejecting -ve values, therefore $a = 3, b = 4$



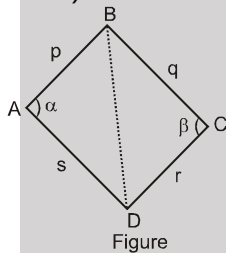
25. Let $f(x) = \frac{\sin x}{x}$
- $$f'(x) = \frac{x \cos x - \sin x}{x^2} = \frac{\cos x(x - \tan x)}{x^2} < 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right); (\because \tan x > x)$$
- $$f''(x) = \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^3}$$
- Let $g(x) = -x^2 \sin x - 2x \cos x + 2 \sin x$
- $$g'(x) = -x^2 \cos x < 0 \quad \forall x \in (0, \pi/2)$$
- for $x > 0$, we have $g(x) < g(0)$ i.e. $g(x) < 0$
- $$\therefore f'(x) < 0 \text{ and } f''(x) < 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow f\left(\frac{A+B+C}{3}\right) > \left(\frac{f(A)+f(B)+f(C)}{3}\right)$$

$$\Rightarrow \frac{\sin\left(\frac{A+B+C}{3}\right)}{\frac{A+B+C}{3}} \geq \left(\frac{\frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C}}{3}\right)$$

$$\Rightarrow \frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C} \leq \frac{9\sqrt{3}}{2\pi}$$

26. Area ($\Delta ABCD$) = Area of ΔADB + Area of ΔBDC



$$A = \frac{1}{2} ps \sin \alpha + \frac{1}{2} qr \sin \beta$$

$$\frac{dA}{d\alpha} = \frac{1}{2} ps(\cos \alpha) + \frac{1}{2} qr \cos \beta \quad \frac{d\beta}{d\alpha} = 0 \Rightarrow \frac{d\beta}{d\alpha} = \frac{-ps \cos \alpha}{qr \cos \beta}$$

$$BD^2 = p^2 + s^2 - 2ps \cos \alpha = q^2 + r^2 - 2qr \cos \beta$$

$$\text{Differentiating we get } -2ps(-\sin \alpha) = -2qr(-\sin \beta) \quad \frac{d\beta}{d\alpha} \Rightarrow \frac{d\beta}{d\alpha} = \frac{ps \sin \alpha}{qr \sin \beta}$$

$$\Rightarrow -\frac{ps \cos \alpha}{qr \cos \beta} = \frac{ps \sin \alpha}{qr \sin \beta} \Rightarrow \sin \alpha \cos \beta + \cos \alpha \sin \beta = 0 \Rightarrow \sin(\alpha + \beta) = 0 \Rightarrow \alpha + \beta = \pi$$

$$\text{Also, } \frac{dA}{d\beta} = \frac{1}{2} ps \frac{\sin(\alpha + \beta)}{\sin \beta} = 0 \Rightarrow \alpha + \beta = \pi$$

$$\text{If } \alpha + \beta < \pi \text{ then } \frac{dA}{d\beta} > 0$$

$$\text{If } \alpha + \beta > \pi, \text{ then } \frac{dA}{d\beta} < 0$$

$\therefore$ By 1st derivative test A has maxima when $\alpha + \beta = \pi \Rightarrow A, B, C, D$ are concyclic



27. Let $u = 2^x + 2^{-x}$

$$u^3 = 8^x + 8^{-x} + 3(2^x)(2^{-x})(2^x + 2^{-x}) \Rightarrow u^3 - 3u = 8^x + 8^{-x}$$

$$\text{also, } 4^x + 4^{-x} = u^2 - 2 \Rightarrow f(x) = u^3 - 3u - 4(u^2 - 2) = u^3 - 4u^2 - 3u + 8$$

$$\text{Let } g(u) = u^3 - 4u^2 - 3u + 8 ; u \geq 2$$

$$g'(u) = 3u^2 - 8u - 3 = (3u + 1)(u - 3)$$

$$\text{putting } g'(u) = 0 ; \text{ we get } u = 3$$

$$g''(u) = 6u - 8 \Rightarrow g''(3) = 10 > 0$$

$$\Rightarrow u = 3 \text{ is point of minima} \Rightarrow g(3) = 27 - 36 - 9 + 8 = -10 \Rightarrow \text{minimum } f(x) = -10$$

28. Let $f(x) = \frac{\log_e(2x-1)}{\log_e x}$ for $x > 1$

$$\text{Now } f'(x) = \frac{(2x-1) \log_e(2x-1) - 2x \log_e x}{x(2x-1) \{\log_e(2x-1)\}^2}$$

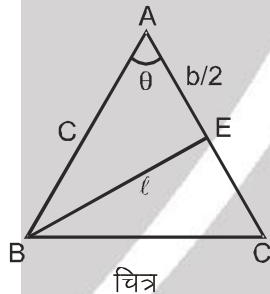
$$\text{Let } g(x) = (2x-1) \log_e(2x-1) - 2x \log_e x$$

$$\Rightarrow g'(x) = 2 \log_e(2x-1) - 2 \log_e x + 2 - 2 = 2 \log_e \left(2 - \frac{1}{x}\right) > 0 \text{ for } x > 1$$

$$\Rightarrow \text{for } x > 1, \text{ we have } g(x) > g(1) \Rightarrow g(x) > 0 \Rightarrow f'(x) > 0 \text{ for } x > 1 \Rightarrow f(x) \text{ is increasing for } x > 1$$

$$4 > 3 > 2 \Rightarrow f(4) > f(3) > f(2) \Rightarrow \frac{\log_e 7}{\log_e 4} > \frac{\log_e 5}{\log_e 3} > \frac{\log_e 3}{\log_e 2}$$

29. $\Delta = \frac{1}{2} bc \sin \theta = \frac{1}{2} c^2 \sin \theta (\because b = c)$



$$\text{In } \Delta ABE, \text{ using cosine rule, } \ell^2 = c^2 + \frac{b^2}{4} - bc \cos \theta = \frac{5c^2}{4} - c^2 \cos \theta$$

$$c^2 = \frac{4\ell^2}{5 - 4\cos \theta}$$

$$\therefore \Delta = 2\ell^2 \cdot \frac{\sin \theta}{5 - 4\cos \theta}$$

$$\frac{+}{-} \cos^{-1} 0.8$$

$$\frac{d\Delta}{d\theta} = \frac{2\ell^2 \cdot ((5 - 4\cos \theta)\cos \theta - \sin \theta (4\sin \theta))}{(5 - 4\sin \theta)^2} \quad \text{signs of } \frac{d\Delta}{d\theta} = \frac{2\ell^2 \cdot (5\cos \theta - 4)}{(5 - 4\sin \theta)^2}$$

$$\therefore \text{For } \Delta \text{ to be maximum, } \cos \theta = 0.8$$



30. $y = 1 - x^2$

Consider point P $(x_0, 1 - x_0^2)$ $0 < x_0 \leq 1$

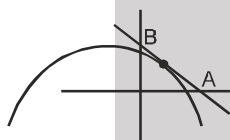
equation of tangent at P is $y - (1 - x_0^2) = -2x_0(x - x_0)$

intersection with x-axis at $\Rightarrow A \equiv \left(x_0 + \frac{(1 - x_0^2)}{2x_0}, 0 \right)$

intersection with y-axis at $B(0, 2x_0^2 + (1 - x_0^2))$ area of ΔOAB $\Delta = \frac{1}{2} \frac{(x_0^2 + 1)^2}{2x_0} = \frac{1}{4} \frac{(x_0^2 + 1)^2}{x_0}$

$$\frac{dA}{dx_0} = \frac{(x_0^2 + 1)}{4x_0^2} [3x_0^2 - 1] = 0 \quad \Rightarrow x_0 = \frac{1}{\sqrt{3}}$$

$\frac{dA}{dx_0}$ changes sign from +ve to -ve



Figure

at $x_0 = \frac{1}{\sqrt{3}}$ So point of minimum $\Rightarrow A_{\min} = \frac{4\sqrt{3}}{9}$

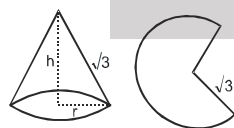
31. $3 = h^2 + r^2$

$$\Rightarrow r^2 = 3 - h^2$$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (3 - h^2) h$$

$$\frac{dV}{dh} = \frac{1}{3} \pi (3 - 3h^2)$$

$$\frac{dV}{dh} = 0 \quad \text{at } h = 1$$



Figure

$$\frac{d^2V}{dh^2} < 0 \quad \text{at } h = 1$$

$$\Rightarrow V_{\max} = \frac{2\pi}{3}$$



32. $V = k \sqrt{\left(\frac{\lambda}{a}\right) + \left(\frac{a}{\lambda}\right)}$

V will be minimum when $\frac{\lambda}{a} + \frac{a}{\lambda}$ will be minimum

$$A.M. \geq G.M.$$

$$\frac{\frac{\lambda}{a} + \frac{a}{\lambda}}{2} \geq \sqrt{\frac{\lambda}{a} \times \frac{a}{\lambda}}$$

$$\frac{\lambda}{a} + \frac{a}{\lambda} \geq 2$$

$$\Rightarrow \text{minimum of } \frac{\lambda}{a} + \frac{a}{\lambda} = 2$$

$$V_{\min} = K\sqrt{2} \text{ which is independent of } a.$$

33. $P \equiv (R \cos\theta, R \sin\theta) \Rightarrow a(R \cos\theta)^2 + 2b(R \cos\theta)(R \sin\theta) + a(R \sin\theta)^2 = c$

$$R^2 [a \cos^2\theta + b \sin 2\theta + a \sin^2\theta] = c \Rightarrow R^2 = \frac{c}{a + b \sin 2\theta}$$

for minimum distance $\sin 2\theta = 1$

$$\therefore R^2 = \frac{c}{a+b} \Rightarrow R = \sqrt{\frac{c}{a+b}}$$

34. Let $\phi(x) = x + \sqrt{1+x^2}$

$$\phi'(x) = 1 + \frac{x}{\sqrt{1+x^2}}$$

If $x < 0$, $-|x| = x$

$$\phi'(x) = \frac{\sqrt{1+x^2} - |x|}{\sqrt{1+x^2}} = \frac{\sqrt{1+x^2} - \sqrt{x^2}}{\sqrt{1+x^2}} > 0$$

If $x > 0$, $\phi'(x) > 0$

Hence $\phi(x)$ is increasing

As we know $e^x \geq x + 1 \Rightarrow \phi(e^x) \geq \phi(x + 1)$

$$e^x + \sqrt{1+e^{2x}} \geq x + 1 + \sqrt{1+(x+1)^2}$$



35. Let $x > -1$

Consider $f(x) = (1+x)\ln(1+x) - \tan^{-1} x$

$$f'(x) = \ln(1+x) + 1 - \frac{1}{1+x^2}$$

$$f''(x) = \frac{(1+x)^2 + 3x^2 + x^4}{(1+x)(1+x^2)^2} > 0 \Rightarrow f'(x) \text{ is increasing}$$

For $x < 0$, $f'(x) < f'(0) \Rightarrow f'(x) < 0 \Rightarrow f(x) \text{ is decreasing} \Rightarrow f(x) > f(0) \Rightarrow f(x) > 0$

$$\Rightarrow (1+x)\ln(1+x) - \tan^{-1} x > 0$$

$$\ln(1+x) > \frac{\tan^{-1} x}{x+1}$$

For $x > 0$, $f'(x) > f'(0) \Rightarrow f'(x) > 0 \Rightarrow f(x) \text{ is increasing} \Rightarrow f(x) > f(0) \Rightarrow f(x) > 0 \Rightarrow \ln(1+x) > \frac{\tan^{-1} x}{x+1}$

Hence larger of these is $\ln(1+x)$.

36. $\cos x \in (0, 1) \Rightarrow f'(\cos x) < 0, f''(\cos x) > 0$

$$g'(x) = \sin x \cos x \left(\frac{f'(\sin x)}{\sin x} - \frac{f'(\cos x)}{\cos x} \right)$$

$$\text{Consider } \phi(t) = \frac{f'(t)}{t}, t \in (0, 1)$$

$$\phi'(t) = \frac{f''(t)t - f'(t)}{t^2}$$

$$\therefore f''(\sin x) > 0 \Rightarrow f''(t) > 0$$

$$f'(\sin x) < 0 \Rightarrow f'(t) < 0 \Rightarrow \phi'(t) > 0 \quad t \in (0, 1)$$

$$\phi(t) \text{ is increasing. For } x \in \left(0, \frac{\pi}{4}\right) \cos x > \sin x$$

$$\phi(\cos x) > \phi(\sin x) \quad \frac{f'(\cos x)}{\cos x} > \frac{f'(\sin x)}{\sin x}$$

$$\Rightarrow g'(x) < 0 \Rightarrow g(x) \text{ is decreasing. Similarly, for } x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) g(x) \text{ is increasing}$$

37. Let $P(x) = x^{2013} - x^{2012} - 1007x^2 + 1007x + k$

as $P(x)$ is a polynomial function in x hence it is everywhere continuous and differentiable

$$\text{also } P(0) = 0 + k = k; P(1007^{1/2011}) = k$$

hence by Rolle's theorem $P'(x) = f(x) = 0$

for atleast one real value of ' x ' in given interval.

39. Clearly $g(x)$ satisfies condition in LMVT $\Rightarrow \frac{g(5) - g(0)}{5 - 0} = g'(c), c \in (0, 5) \quad \frac{f(5) - f(0)}{5} = g'(c) - \frac{5}{6} = g'(c)$

39. Let two consecutive zero of $f(x)$ be a and b $f(a) = 0 = f(b)$.





If possible, suppose $g(x)$ has no zero. Define $\phi(x) = \frac{f(x)}{g(x)}$

$\phi(x)$ satisfies conditions in Rolle's theorem, $\Rightarrow \phi'(c) = 0$ for at least one $c \in (a, b)$

$$\Rightarrow f'(c) g(c) - f(c) g'(c) = 0$$

Which is a contradiction to given condition $f(x) g'(x) \neq f'(x) g(x)$

Hence our supposition that $g(x)$ has no zero is wrong $\Rightarrow g(x)$ has at least one zero.

40. Let $f'(x) = \phi'(x+a) - \phi'(x) \Rightarrow f(x) = \phi(x+a) - \phi(x) + k$

$$f(0) = \phi(a) - \phi(0) + k$$

$$f(2a) = \phi(3a) - \phi(2a) + k \Rightarrow f(0) = f(2a)$$

By Rolle's theorem on $[0, 2a]$, $f'(c) = 0$ for at least one $c \in (0, 2a)$

$$\Rightarrow \phi'(x+a) = \phi'(x) \text{ has at least one root in } (0, 2a)$$

41. $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$

$$f'(x) = 8a - 6a \cos 6x - 7 - 5 \cos 5x = 8a - 7 - 6a \cos 6x - 5 \cos 5x$$

$f(x)$ is an increasing function

$$f'(x) > 0$$

$$\therefore 8a - 7 > 6a + 5$$

$$\Rightarrow 2a > 12$$

$$a > 6$$

$$a \in (6, \infty)$$

42. $g'(x) = \frac{h'(x)}{h(x)} \quad g''(x) = \frac{h(x) \cdot h''(x) - (h'(x))^2}{(h(x))^2}$

$$\Rightarrow g''(x) < 0$$

$\therefore g(x)$ is concave down on J .

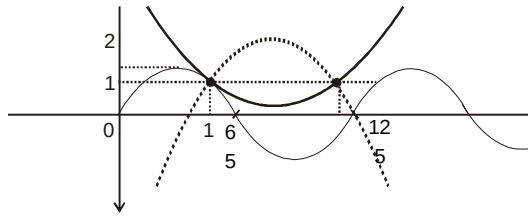
43. $f''(x) = 2ax + 2(a+2)$

$$\text{Point of inflection is } x = -\frac{a+2}{a}$$

$$-\frac{a+2}{a} < 0 \Rightarrow a \in (-\infty, -2) \cup (0, \infty)$$

44. Period of $y = 2 \sin \frac{5\pi}{6} x$ is $\frac{2\pi}{5\pi/6} = \frac{12}{5}$ & $y = \alpha(x-1)(x-2) + 1$ which is a quadratic. So for given

information $(1, 1)$ is the common point to two curves and the possible graph would be



α can not be negative α must be positive for these two graphs to touch each other at (1, 1)

(Which can be the only possible point of contact) $\frac{dy}{dx}$ of both curves must be same.

$$y = 2\sin \frac{5\pi}{6}x \Rightarrow \frac{dy}{dx} = \left(2\cos \frac{5\pi}{6}x\right) \cdot \frac{5\pi}{6}$$

$$\left(\frac{dy}{dx}\right)_{\text{at } x=1} = \left(2\cos \frac{5\pi}{6}\right) \cdot \frac{5\pi}{6} = \frac{5\pi}{6} \cdot \cos \left(\pi - \frac{\pi}{6}\right) = -\frac{5\pi}{3} \cdot \frac{\sqrt{3}}{2} \quad \dots(1)$$

$$y = \alpha x^2 - 3\alpha x + 2\alpha + 1 \Rightarrow \frac{dy}{dx} = 2\alpha x - 3\alpha$$

$$\left(\frac{dy}{dx}\right)_{\text{at } x=1} = 2\alpha - 3\alpha = -\alpha \quad \dots(2)$$

$$\text{for (1) \& (2) } \alpha = \frac{5\pi}{2\sqrt{3}} \Rightarrow \frac{\sqrt{3}\alpha}{5\pi} = \frac{1}{2}$$

45. Let d be distance between $(k, 0)$ and any point (x, y) on curve.

$$d = \sqrt{(k-x)^2 + y^2}$$

$$d = \sqrt{-x^2 + 2(1-k)x + k^2} \quad (\because y^2 = 2x - 2x^2)$$

$$\text{Maximum } d = \sqrt{\frac{4(-1)k^2 - 4(1-k)^2}{4(-1)}} \quad \text{Maximum } d = \sqrt{2k^2 - 2k + 1}$$

46. $f(x) = (x+1)(px^2 + (q-p)x + p)$

$$g(x) = px^2 + (q-p)x + p$$

$$g(2) = 3p + 2q < 0$$

$$g(3) = 7p + 3q > 0$$

$\Rightarrow$ one root is -1 , one root lies between $(2, 3)$, one root lies between $\left(\frac{1}{3}, \frac{1}{2}\right)$

47.

